Impact of Medicaid Pregnancy Dental Benefits on Prenatal Dental Utilization and Birth Outcomes*

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Abstract

Despite elevated risks for oral health problems, more than half of pregnant women do not visit the dentist during pregnancy. The Medicaid program covers more than 40% of US births and represents an important source of pregnancy dental coverage. We examine the effects of expanding state-level pregnancy dental benefits using a difference-in-differences approach and data from the 2012-2019 Pregnancy Risk Assessment Monitoring System (PRAMS). We find that providing dental coverage to pregnant Medicaid recipients increases dental cleaning rates by 9.5 percentage points, or 37% relative to baseline. We also examine linked birth certificate data given evidence that poor oral health during pregnancy is associated with adverse pregnancy and birth outcomes. We find evidence of reductions in poor infant health outcomes following expansion of dental benefits, though estimates vary somewhat across specifications. An analysis of mechanisms suggests that Medicaid pregnancy dental benefits increase rates of pregnancy dental insurance, increase receipt of dental counseling, and improve self-reported general health status.

JEL Codes: I13, I14, I18, J13

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1 Introduction

Poor oral health during pregnancy may have long term implications for the oral and overall health of expectant mothers and their children (Dye et al., 2011; Ide & Papapanou, 2013; Jeffcoat et al., 2014; Chaffee et al., 2014; Iheozor-Ejiofor et al., 2017; Nasseh et al., 2017). Oral health problems can cause pain that interferes with eating, speaking, and overall quality of life (Naito et al., 2006; CDC, 2024a). Moreover, poor oral health has been linked to systemic health conditions including cardiovascular disease (Oliveira et al., 2010; Sanz et al., 2020), diabetes (Díaz-Romero et al., 2005; Chee et al., 2013; Nasseh et al., 2017), respiratory conditions (Manger et al., 2017), and adverse pregnancy and birth outcomes (Jeffcoat et al., 2014; Puertas et al., 2018). More than one-third of pregnant women report experiencing dental problems (Lee et al., 2022b). Despite elevated risks for oral health problems and the availability of effective interventions to address these issues, more than half of pregnant people do not visit the dentist during pregnancy (Kranz & Estrada-Darley, 2022; Lee et al., 2022b). We leverage plausibly exogenous variation in state-level Medicaid pregnancy dental benefits to investigate the extent to which reducing costs incentivizes prenatal dental services use and translates to improvements in infant health.

Medicaid finances more than 40% of US births and therefore represents an important lever for improving maternal and infant health outcomes (KFF, 2023). Federal law requires coverage of certain services such as hospital and prenatal care, but there is no federally-set minimum coverage requirement for pregnancy dental services. All states currently offer emergency dental services to pregnant enrollees (i.e., tooth extractions), and most states offer some additional non-emergency services (Ranji et al., 2022). Coverage of preventive and restorative services such as dental cleanings and fillings has varied over time with at least 12 states adding these benefits since 2012.[1] As an optional benefit, states may also elect to reduce pregnancy dental benefits given budgetary challenges or changes in leadership priorities. State-level changes in pregnancy dental benefits provide an opportunity to study how Medicaid program benefit design influences maternal health behaviors and perinatal outcomes.

In this paper, we examine the effects of Medicaid pregnancy dental benefits using 2012-2019 data from the Pregnancy Risk Assessment Monitoring System (PRAMS). Our main analysis uses a stacked difference-in-differences approach to account for variation in policy timing and an enhanced dataset compiled via direct state data release requests. We match the month and year of policy implementation with birth dates to define policy exposure. Our primary outcome is reported receipt of a dental cleaning during pregnancy. Given evidence that poor oral health is associated with worse birth outcomes, we also

[1]Based on authors' review of multiple sources including KFF, CMS State Plan Amendments, MACPAC, and the National Academy for State Health Policy.

use linked birth certificate data to examine preterm birth, small for gestational age (10th percentile), low birth weight (under 2500 grams), and very low birth weight (under 1500 grams).

We find that exposure to Medicaid dental benefits during pregnancy significantly increases prenatal dental cleaning rates by 9.5 percentage points, representing approximately a 37 percent increase relative to the baseline rate of 25.5 percent. We also find evidence of improvements in birth outcomes. In our main specification using wild cluster bootstrap inference, we estimate reductions in preterm births (1.35 percentage point decrease, $p<0.05$), low birth weight (0.59 percentage point decrease, $p<0.10$), and very low birth weight (0.47 percentage point decrease, $p<0.10$). We also estimate a 2.19 percentage point decrease in small for gestational age that is not statistically significant at conventional levels, though event study analyses suggest a pattern of decline following pregnancy dental benefit expansions. Subgroup analyses indicate that increases in dental cleanings are observed across most demographic groups with between-group differences generally statistically indistinguishable from zero.

In terms of mechanisms, we find that Medicaid pregnancy dental benefits increase rates of dental insurance by 13.5 percentage points. Furthermore, women who gain pregnancy dental benefits are more likely to report receiving counseling on how to care for their teeth and gums. We also investigate potential spillovers to broader maternal health behaviors. We find no evidence of changes in prenatal care use. Point estimates for smoking during pregnancy are negative, indicating about a 1.49 percentage point decline, but statistically indistinguishable from zero.

Event study analyses generally do not indicate a violation of the parallel trends assumption for most outcomes. Results are also generally robust across multiple sensitivity checks such as varying the stacked event window, including births during the COVID-19 pandemic, excluding each treatment state in turn, and alternative ways of defining the sample. However, in some instances, results for some birth outcomes are less precise.

While states can offer different benefit packages to pregnant and non-pregnant populations, several treatment states in our analysis adopted dental benefits for these groups simultaneously. In an extension to our main analysis, we address this potential confounding and explore the role of pre-conception dental benefits in two ways. First, we restrict our analysis to treatment states that offered dental benefits to both pregnant and non-pregnant adults and examine income-based subsamples of women more and less likely to have been eligible for Medicaid before pregnancy. Second, we augment our main analysis by including a control for exposure to pre-conception dental benefits. Across these specifications, the estimated effects of pregnancy dental benefits on prenatal dental cleaning remain stable, and we continue to observe reductions in adverse birth outcomes,

although some estimates are less precise. Taken together, these findings indicate that our main results are unlikely to be driven by concurrent exposure to non-pregnancy dental benefits and support an independent role for pregnancy-specific dental coverage in improving maternal oral health and infant outcomes.

We contribute new multi-state evidence of the effects of Medicaid pregnancy dental benefits using a rigorous difference-in-differences design, extensive robustness checks, and detailed maternal survey data. A number of quasi-experimental analyses have examined the effects of Medicaid dental benefits, finding increases in dental visits and improvements in oral health (for example, Choi (2011); Decker & Lipton (2015); Semprini et al. (2024)), reductions in emergency department visits (Singhal et al., 2015; Elani et al., 2020a; Ranade et al., 2025), and supply-side responses (Buchmueller et al., 2016; Huh, 2021). Other research, while less extensive, has also examined the impacts of other optional Medicaid benefits such as routine vision, smoking cessation, and abortion services (Lipton & Decker, 2015; Kostova et al., 2018; Kim et al., 2025). Despite the clinical importance of oral health during pregnancy, quasi-experimental evidence on Medicaid pregnancy dental benefits remains limited. We are aware of only one prior study that evaluates such benefits in a single state (Naavaal & Harless, 2022). Moreover, little is known about how these policies affect birth outcomes.

Our study advances the literature by moving beyond dental utilization to examine birth outcomes and by providing plausible evidence on the mechanisms through which these effects may operate. More broadly, research has shown that expanding Medicaid eligibility and program generosity increase coverage and prenatal care use during pregnancy, whereas evidence on infant health outcomes remains mixed (Currie & Gruber, 1996; Howell, 2001; Sonchak, 2015; Wherry et al., 2017; Wherry, 2018; Saygili & Bayindir, 2024). This paper contributes to this line of research by focusing on access to a specific type of prenatal care—dental benefits during pregnancy.

The rest of the paper proceeds as follows. Section 2 reviews the related literature and discusses potential mechanisms. Section 3 describes the PRAMS data and other supplemental datasets, and details the construction of the main variables and summary statistics. Section 4 outlines the empirical specification. Section 5 presents the main results, and Section 6 concludes.

2 Background and Mechanisms

2.1 Pregnancy and Oral Health

In a national consensus statement, the Oral Health Care During Pregnancy Expert Workgroup recommended that pregnant people receive prenatal dental care, but pregnant people are less likely to visit the dentist (44%) than reproductive-age people who are not

pregnant (65%) (CareQuest Institute for Oral Health, 2023). Moreover, non-Hispanic Black women are less likely to visit the dentist during pregnancy than non-Hispanic White women. Medicaid-enrolled pregnant people are also less likely to visit the dentist than those with private insurance (Lee et al., 2022b).

Oral health problems are common among pregnant people (CDC, 2024b). Hormone fluctuations during pregnancy increase the risk of developing gingivitis, a mild form of gum disease affecting up to three-quarters of pregnant people (Figuero et al., 2013; CDC, 2024b). If left untreated, gingivitis can progress to periodontitis, causing damage to the gums and bone that may lead to tooth loss. Pregnant people are also more susceptible to tooth decay given changes in eating and oral hygiene habits (e.g., reduced tooth brushing and flossing) (Boggess et al., 2010; Hunter & Yount, 2011). Poor oral hygiene and the presence of gum disease during early pregnancy are highly predictive of gum disease in later pregnancy (Gil-Montoya et al., 2023). Moreover, intervention studies suggest nonsurgical periodontal therapy is highly effective in resolving symptoms among pregnant people (Michalowicz et al., 2006; Kaur et al., 2014; Bobetsis et al., 2020).

Poor oral health during pregnancy may increase inflammation, thereby worsening systemic health and negatively affecting the fetal environment (Offenbacher et al., 2006; Horton et al., 2008; Madianos et al., 2013; Figuero et al., 2020). Furthermore, periodontal pathogens have been identified in the placenta and amniotic fluid, allowing for the possibility of direct impacts of maternal oral infection on infant health (Bobetsis et al., 2020). Despite a large body of research examining these associations, evidence on the relationship between maternal oral health and infant health outcomes remains mixed and uncertain.

Most observational studies find that maternal periodontal disease is associated with higher risks of preterm delivery and low birth weight (de Oliveira et al., 2021; Uwambaye et al., 2021; Lee et al., 2022c; Castaño-Suárez et al., 2024). Moreover, some evidence finds that receiving routine dental care during pregnancy may improve perinatal outcomes (Lee et al., 2025). Magnitudes vary with some research indicating that the odds of preterm birth are as much as two to six times higher among women with periodontitis compared to those without, while other findings are significant but more modest (Uwambaye et al., 2021; de Oliveira et al., 2021; Lee et al., 2022c). Retrospective cohort studies using linked medical and dental claims also indicate that treatment for periodontal disease at baseline precedes lower healthcare costs, including a large, 74% reduction in spending on atypical pregnancy care (Jeffcoat et al., 2014; Nasseh et al., 2017).

By contrast, results from rigorous randomized controlled trials (RCTs) of periodontal interventions during pregnancy are mixed. A recent systematic review and meta-analysis concluded that periodontal intervention was associated with reductions in the inflammatory marker C-reactive protein and preterm deliveries with no significant

impacts on other birth outcomes (Orlandi et al., 2022). Overall, periodontal intervention reduced the risk of preterm birth by 23% (Orlandi et al., 2022). However, high-quality multi-center RCT evidence has generally failed to find an impact (Michalowicz et al., 2006; Bobetsis et al., 2020). Periodontal interventions generally occurred during the second or third trimester of pregnancy in this research. We are not aware of RCT evidence where the intervention occurred before conception or during the first trimester, which leaves open the possibility that earlier treatment may be more effective. Furthermore, impacts may differ by population characteristics including socioeconomic status and geographic region.

Our analysis complements this literature by examining the effects of Medicaid pregnancy dental benefits in a large, geographically diverse, Medicaid-covered population using quasi-experimental policy variation. Rather than evaluating specific clinical periodontal interventions, we study the impacts of insurance coverage expansions that plausibly affect the timing, intensity, and continuity of dental care during pregnancy. In extensions to our main analysis, we further assess the role of exposure to pre-conception Medicaid dental benefits, providing indirect evidence on the role of timing and prior access to care.

2.2 Medicaid Dental Benefits

Medicaid is the single largest health insurer in the United States covering 70.5 million people as of June 2025 (KFF, 2025). All states must provide preventive and restorative dental services to enrollees up to age 20 under the Early and Periodic, Screening, Diagnostic, and Treatment benefit (Centers for Medicare & Medicaid Services, 2024). However, dental services for adults ages 21 and older are an optional benefit provided at the state's discretion.[2] States may also opt to provide different dental benefit packages to pregnant and non-pregnant adults. Overall, many states provide more generous dental benefits during pregnancy (National Academy for State Health Policy, 2022).

Recent correlational research suggests that Medicaid dental coverage is associated with increases in dental care utilization among pregnant people (Robison et al., 2021; Lee et al., 2022a). One cross-sectional analysis found that Medicaid-enrolled pregnant people with no Medicaid dental coverage were less likely to have a dental cleaning during pregnancy (27%) when compared to those with limited benefits (e.g., fewer than 100 services or an annual spending cap of less than $1,000) (37%) and extensive benefits (e.g., at least 100 services and an annual spending cap of at least $1,000) (45%) (Lee et al., 2022a). The only quasi-experimental analysis we are aware of examined Virginia's 2015

[2]Federal law generally requires that managed care enrollees have access to at least the same level of benefits as people enrolled in the state's fee-for-service program, including optional benefits (MACPAC, 2025). Therefore, state-required dental benefits would be likely to increase dental coverage rates among both fee-for-service and managed care enrollees.

addition of a comprehensive Medicaid pregnancy dental benefit and found that reports of receiving a dental cleaning increased among Medicaid-enrolled pregnant people relative to privately insured pregnant people before and after the policy's implementation (Naavaal & Harless, 2022). However, this research focused on a single state and included limited pre-policy implementation data. While not examining Medicaid dental benefits directly, one analysis found that Medicaid eligibility expansions to pregnant people and infants during their first year of life were associated with a significant reduction in permanent tooth loss on reaching young adulthood for non-Hispanic Black cohorts (Lipton et al., 2016). These findings indicate potential improvements in access to dental care among pregnant populations and children and also provide support for the notion of longer-run oral health improvements following early life interventions.

While evidence on the impacts of Medicaid pregnancy dental benefits remains somewhat limited, there is a large body of research on dental benefits for non-pregnant adults. Quasi-experimental estimates of the impacts of Medicaid dental benefits on dental visits range from 6-14 percentage points, (for example, Choi (2011); Decker & Lipton (2015); Abdus & Decker (2019); Lipton (2021)). Moreover, other researchers have found that Medicaid dental benefits reduce out-of-pocket spending on dental care (Abdus & Decker, 2019), reduce emergency department visits for oral conditions (Singhal et al., 2015; Elani et al., 2020a, 2021; Ranade et al., 2025), and that they induce a supply-side response with dentists increasing their acceptance of Medicaid patients and relocating to counties with larger Medicaid-insured populations (Buchmueller et al., 2016; Huh, 2021). Finally, Medicaid adult dental benefits have been shown to have spillover effects on children's dental care use with effects that appeared to be concentrated among those with parents who visited a dentist (Lipton, 2021; Hill et al., 2022).

2.3 Mechanisms

The likely primary mechanism for an impact of Medicaid pregnancy dental benefits on dental visits is via a reduction in out-of-pocket costs for pregnant enrollees. According to published fee schedules, the average cost of a dental cleaning ranges from $90 to $120, and restorative services such as composite fillings and crowns typically range from $100 to over $1,000 per tooth (Carefree Dental, 2021; American Dental Care Network, 2024). In states that provide Medicaid dental benefits, enrollee cost-sharing amounts are typically small, often between $1 and $3, representing a substantial reduction in enrollee costs for dental services (Kaiser Family Foundation, 2025). Estimates suggest that providing Medicaid dental benefits to non-pregnant adults results in a reduction in annual out-of-pocket costs of $18.88 on average and $179.28 among those with a dental visit (Abdus & Decker, 2019). In addition, Medicaid dental benefits for non-pregnant adults reduce reports of needing but not receiving dental care due to cost by more than 12 percentage points (Decker &

Lipton, 2015).

While we cannot directly observe out-of-pocket costs in PRAMS, we find that Medicaid pregnancy dental benefits increase dental insurance rates by 13.5 percentage points. This result suggests both that pregnant enrollees are aware that they have Medicaid dental benefits and that many would not otherwise have another source of private dental insurance. Given low cost sharing in Medicaid, enrollees without another source of dental insurance are likely to experience a large reduction in out-of-pocket costs for dental services when pregnancy dental benefits are offered. Furthermore, out-of-pocket costs would likely decline for some enrollees who would have otherwise had private dental insurance and gained Medicaid dental coverage, though we are not able to measure the scope of changes in dental coverage sources.

Medicaid pregnancy dental benefits could also raise awareness of the importance of oral health during pregnancy through outreach or beneficiary communication. However, we do not find evidence of this possibility in the PRAMS.

In addition to demand-side effects, providing Medicaid dental benefits could also have supply-side impacts because of changing patient mix and lower Medicaid reimbursement rates relative to private dental insurance (Gupta et al., 2017). Research indicates that when states offer Medicaid dental benefits, dentists are more likely to participate in Medicaid and provide more services to Medicaid-insured patients (Buchmueller et al., 2016). Furthermore, Medicaid dental benefits induce dentists to relocate to counties with higher Medicaid-eligible populations (Huh, 2021). Therefore, existing evidence indicates a positive supply-side response to accommodate increased demand when states add Medicaid dental benefits.

Our finding that Medicaid pregnancy dental benefits increase dental visits raises the possibility of improvements in oral health among pregnant women. In addition to an increase in dental cleaning rates, we also find a significant 5.8 percentage point increase in reports of receiving advice on how to care for teeth and gums. Thus, Medicaid pregnancy dental benefits may improve oral health both via care received during a dental visit and also through improvements in caring for teeth and gums at home. Previous research focusing on non-pregnant Medicaid enrollees indicates that Medicaid dental benefits reduce untreated tooth decay by about 9.5 percentage points (Decker & Lipton, 2015). Moreover, researchers found that the Affordable Care Act's Medicaid expansion reduced untreated tooth decay by 16.8 percentage points in expansion states that offered dental benefits (Elani et al., 2021).

There are biologically plausible pathways through which treating oral health problems could improve birth outcomes (e.g., via reducing inflammation and exposure to oral bacteria), although the evidence to date is mixed, as described. While we are limited in our ability to examine this pathway in the PRAMS, recent research suggests that dental

benefits for non-pregnant adults are associated with improvements in general reported health status, which may serve as a proxy for systemic health (Wehby et al., 2025). We also confirm this finding among reproductive-aged women using data from the Behavioral Risk Factor Surveillance System.[3]

In addition to possible impacts on maternal systemic health and the fetal environment, pregnancy dental benefits could have indirect impacts on infant health by changing maternal health behaviors. Other types of prenatal care use could increase if dentists counsel pregnant patients on current guidelines for obstetric visits. Conversely, dental visits may crowd out other prenatal care use given time and work constraints. We examine the effects of pregnancy dental benefits on the Kotelchuk index, a measure of prenatal care adequacy, and find no evidence of an effect. Pregnancy dental benefits could also affect other types of health behaviors if dentists counsel pregnant women on dietary or smoking habits, via an income effect, or moral hazard. Given the adverse consequences of smoking during pregnancy, we assess whether pregnancy dental visits affect smoking prevalence. While point estimates for smoking during pregnancy are negative, they are not statistically significant at conventional levels in our main specification using wild cluster bootstrap inference. We note that there is evidence that Medicaid eligibility expansions to pregnant women during the 1980s and 1990s increased smoking behavior (Dave et al., 2019). We study a more recent time period and the addition of a specific type of coverage, which may explain this difference.

Overall, our analysis of mechanisms indicates that an increase in dental insurance rates and associated reductions in out-of-pocket costs is likely the primary driver of increased dental visits. In terms of improvements in birth outcomes, our analysis suggests that pregnancy dental benefits are associated with increased rates of counseling on caring for teeth and gums at home, which may complement in-office care in improving oral health. We also find that Medicaid dental benefits improve self-reported general health status among reproductive-age women, which may proxy for improvements in systemic health.

3 Data

The data for this study are drawn from the Pregnancy Risk Assessment Monitoring System (PRAMS), a state-based surveillance system managed by the Centers for Disease Control and Prevention (CDC) in collaboration with state health departments. PRAMS collects comprehensive, population-based data on maternal experiences before, during, and shortly after pregnancy, with an emphasis on maternal health behaviors and access to care. The survey samples a state-representative subset of women who have recently given birth, sampling between 1,000 and 3,000 individuals annually from each participating

[3]We examine reproductive-aged women rather than pregnant women because an analysis of pregnant women in BRFSS would be under-powered in our context.

site (typically a state). The dataset includes self-reported measures of dental care access during pregnancy and linked birth certificate information on birth outcomes.

PRAMS data availability varies by year and state, as the CDC only releases data for a given state-year if the site meets a minimum weighted response rate threshold (typically 50%). Additionally, even when a state meets the threshold, its data may not be publicly released unless the site opts to make it available. To ensure more complete coverage, we submitted site-specific data requests to obtain restricted-use PRAMS data for states not included in the standard release files. As a result, our dataset includes restricted-use data from Colorado (2014), Tennessee (2017-2018), South Carolina (2012-2019), Mississippi (2016-2017), Virginia (2012-2014), Louisiana (2012-2014) and West Virginia (2019). In these cases, South Carolina met response thresholds but did not publicly release their data. Other states fell below the CDC's cutoff in certain years, though typically maintained response rates above 40%. Including these restricted-use files enhances the scope and representativeness of our analysis by reducing bias due to selective state participation.

Our primary outcome is an indicator of having received a dental cleaning at any time during the most recent pregnancy. Furthermore, we investigate four main birth outcomes, including small for gestational age (at or below 10th percentile), preterm birth (before 37 weeks gestation), low birth weight (under 2500 grams), and very low birth weight (under 1500 grams). The PRAMS also provides a rich set of individual and socioeconomic characteristics at the individual level, allowing us to control for potential factors that may influence dental care utilization and pregnancy health.

Our main analysis uses PRAMS data from 2012 to 2019. We begin our analysis in 2012 because this is the first year when the question about dental cleanings during pregnancy is included in the survey. We end our analysis in 2019 to avoid confounding due to the COVID-19 pandemic given evidence of depressed healthcare utilization, interruptions to prenatal care, and infant health impacts (Ferrara et al., 2023; Lee, 2024; Karasek et al., 2025). However, we include data through 2021 in a sensitivity analysis. Information about dental cleanings and birth outcomes is available in all years of our analysis. We also examine related outcomes as possible mechanisms including dental insurance (2012-2015), awareness of the importance of oral health during pregnancy (2012-2015), receipt of counseling on caring for teeth and gums (2012-2015), the Kotelchuk index of prenatal care adequacy (2012-2019), and smoking during pregnancy (2012-2019). While several of these outcomes are only available during 2012-2015, all treatment states in our main analysis implemented policies during this period, allowing us to generate estimates based on meaningful policy variation.

We focus on individuals aged 21 and older who were enrolled in Medicaid during pregnancy. Medicaid enrollment is identified primarily from the source of payment

for delivery on the birth certificate; for records with missing payer information, we supplement with self-reported Medicaid coverage from PRAMS. This restriction allows us to target populations most likely to be affected by Medicaid dental policy changes since all Medicaid enrollees up to age 20 receive dental coverage under the Early and Periodic, Screening, Diagnostic, and Treatment benefit (Centers for Medicare & Medicaid Services, 2024).

Our primary policy indicator is equal to one for dates of birth during or following the month of policy implementation and zero otherwise. This indicator represents any exposure to Medicaid dental benefits during pregnancy. However, our event study analysis allows us to examine whether treatment effects grow with increasing exposure. We also consider exposure based on imputed conception date in a sensitivity analysis.[4]

We documented state policies during 2012-2021 by cross-verifying information from multiple sources, including KFF's Medicaid Benefits database (KFF, 2025), MACPAC reports and compendia (Medicaid and CHIP Payment and Access Commission, 2021), NASHP's state tracker (National Academy for State Health Policy, 2022), state Medicaid provider manuals, and State Plan Amendments in the CMS portal (Centers for Medicare & Medicaid Services, 2025). Following prior research on non-pregnant adults (Decker & Lipton, 2015; Abdus & Decker, 2019; Elani et al., 2020b), we consider states that provide at least one preventive or restorative service to pregnant women as having pregnancy dental coverage. Since states that provide any preventive or restorative services generally cover dental cleanings including all treatment states in our analysis, this measure is arguably most appropriate for examining dental cleaning rates. Other research has used the American Dental Association's classification, which splits non-emergency adult dental coverage into limited (typically fewer than 100 procedures or an annual expenditure cap below $1,000) and extensive (at least 100 procedures or a cap of $1,000 or more) (Wehby et al., 2019; Lyu et al., 2020; Wehby et al., 2022). While we explored using this definition in our analysis, states classified as offering limited coverage in other sources did not always meet our minimum criteria of covering at least one preventive or restorative service for pregnant women.

As Figure 1 shows, five PRAMS states—including Colorado, Illinois, South Carolina, Utah, and Virginia—began offering pregnancy dental benefits during the study period (shown in green). One state (Oklahoma) dropped pregnancy dental benefits during the study period (shown in yellow).[5] Twenty-eight PRAMS states always offered pregnancy dental benefits (shown in blue), and eight states never offered pregnancy dental benefits throughout the entire study period (shown in red). Two states that had not adopted

[4]We prefer to define exposure based on month and year of birth rather than conception date since it does not require any imputation.

[5]While Oklahoma data was missing from the public use data portal, they provided restricted data in response to our request. However, we did not include OK in our main analysis sample since they were the only state that reversed pregnancy dental benefits during our analysis period.

pregnancy dental benefits by 2019 did so after 2019 (West Virginia in 2021 and Delaware in 2020).[6] States shown in gray either did not participate in PRAMS or had data suppressed due to response rates falling below reporting thresholds and did not supply their data in response to our request. Our final sample includes the five treatment states that adopted pregnancy dental benefits during our study period and eight control states that never offered pregnancy dental benefits during our study period.[7]

(Figure 1 here)

Table 1 presents descriptive statistics of the study population stratified by Medicaid pregnancy dental benefit adoption status. For treatment states, we also stratify estimates by pre- and post-policy implementation. The "pre-policy" period is defined as all births occurring prior to the effective month and year when pregnancy dental benefits were introduced in a given treatment state (i.e., no exposure), while the "post-policy" period includes all births occurring after implementation (i.e., any exposure).

(Table 1 here)

About 27% and 30% of women reported receiving a dental cleaning pre- and post-policy implementation in treatment states compared with about 25% in control states. In treatment states pre-policy, about 9% of births were classified as low birth weight, 1.6% as very low birthweight, 11% as preterm, and 11% as small for gestational age. These baseline rates were relatively similar to those in control states. While treatment state rates were relatively similar pre- and post-policy implementation for most birth outcomes, rates of preterm birth declined by about 1.4 percentage points. The average maternal age ranged from 27 to 28 years across groups, and more than half had less than a college education. The sample was racially and ethnically diverse, with some differences between the treatment and control states (e.g., 41% were non-Hispanic white in treatment states at baseline vs. 48% in control states). More than one-third were married, and nearly three-quarters resided in urban areas. There were also some differences between the treatment and control states in terms of state-level policy and healthcare variables. Most notably, dentist supply was somewhat lower and Medicaid managed care penetration rates were substantially higher in control states (61% vs. 26% in treatment states at baseline). We also examine whether pregnancy dental benefits

[6]West Virginia and Delaware are control states in our main analysis through 2019, but treatment states in our sensitivity analysis including data through 2021.

[7]Since our primary analytical strategy is a stacked difference-in-differences model, we exclude the 28 states with pre-existing dental benefits to avoid confounding arising from continually evolving trends in these "already treated" units, consistent with best practices (Wing et al., 2024).

are systematically associated with changes in sample composition and generally find that associations are statistically indistinguishable from zero (Table A.1).[8]

Since PRAMS does not consistently collect measures of general health status, we draw on data from the Behavioral Risk Factor Surveillance System (BRFSS) to examine systemic health as a potential mechanism. BRFSS is a nationally representative, state-based telephone survey administered by the CDC that collects information on health-related behaviors, chronic conditions, and preventive service use among U.S. adults. We focus on reproductive-aged women rather than pregnant women due to larger sample sizes. Thus, we use BRFSS data to examine the association between Medicaid dental benefits for non-pregnant adults and changes in self-reported general health among reproductive-aged women. Since BRFSS has only recently begun collecting information on Medicaid enrollment, we restrict the sample to women with household incomes below the Medicaid eligibility threshold in their state of residence during the interview year. This analysis provides complementary evidence on whether Medicaid dental benefits may influence broader health and well-being, a potential pathway linking coverage to improved birth outcomes.

4 Empirical Strategy

4.1 Stacked Difference-in-differences Approach

This study employs a stacked difference-in-differences (DiD) design, following recent methodological guidance from Wing et al. (2024). In this approach, we define a series of "sub-experiments," each centered around a unique policy adoption date. For each sub-experiment, the treatment group comprises states implementing pregnancy dental benefits on a given date, and the control group consists of states that did not adopt similar benefits within the event window. This design decomposes staggered adoption into a set of two-by-two DiD comparisons, mitigating the bias that can arise in two-way fixed effects models with heterogeneous treatment timing (de Chaisemartin & D'Haultfœuille, 2020; Goodman-Bacon, 2021; Sun & Abraham, 2021).

We adopt this approach for several reasons articulated by Wing et al. (2024). First, our setting includes a moderate number of treated states, which facilitates clean sub-experiment construction. Second, policy adoption is monotonic in five treatment states- most states either maintained consistent coverage or added but did not remove dental benefits during the study period—making treatment assignment consistent over

[8]We regress each demographic characteristic on a binary pregnancy dental benefit indicator, state-by-year controls, and state and year fixed effects. Overall, most associations are statistically indistinguishable from zero when using our preferred wild cluster bootstrap inference with the exception of Hispanic ethnicity.

time.[9] Third, our main PRAMS data are repeated cross-sections that require survey weights to produce representative estimates. The stacked DiD approach accommodates survey weights and does not require panel tracking of individuals. Taken together, these features make stacked DiD an appropriate choice for our empirical setting.

Table 2 summarizes the states included in each sub-experiment used in our primary analysis. We define event windows of up to 48 months before and after policy adoption. Because our analysis period begins in 2012 and several policy adoptions occur in 2014, sub-experiments generally have shorter pre-policy periods than post-policy periods. However, this approach allows enough time for supply- and demand-side effects to evolve post-policy implementation. For example, Utah's policy adoption in October 2013 forms one sub-experiment with the event window spanning January 2012 through October 2016. The nine states that did not offer pregnancy dental benefits during this window serve as controls. We also consider alternative event windows of 24, 36, and 60 months in sensitivity analyses.

(Table 2 here)

4.2 Specification

To estimate the overall treatment effect, we run separate two-by-two DiD models within each sub-experiment (Table 2), comparing outcomes before and after policy adoption in treated states to changes in control states during the same period. These sub-experiment estimates are then aggregated using stacked weights, which correct for bias due to imbalances in the weighting of treatment and control trends across sub-experiments (Wing et al., 2024). We estimate the following stacked difference-in-differences model:

$$Y_{istd} = \alpha_0 + \omega_1 \text{PregDental}_{istd} + \mathbf{X}_{istd}\boldsymbol{\beta} + \mu_{sd} + \lambda_t + \epsilon_{istd} \tag{1}$$

Where Y_{istd} denotes the outcome of interest (e.g., dental cleaning or birth outcome) for individual i in state s, time t, and sub-experiment d. PregDental_{istd} is a binary indicator equal to one if a respondent had any exposure to pregnancy dental benefits during their pregnancy and zero otherwise. $\mathbf{X}_{istd}$ is a vector of individual- and state-level control variables, μ_{sd} represents state-by-sub-experiment fixed effects, and λ_t represents birth quarter-by-year fixed effects. State-by-sub-experiment fixed effects control for time-invariant state-level differences within each policy comparison group and birth quarter-by-year fixed effects account for common outcome trends between treatment and control states. Individual-level controls include education, age in years, race and ethnicity, marital status, household income,[10] an indicator for having either pre-pregnancy

[9] Only Oklahoma dropped pregnancy dental benefits during the study period and we exclude them from our analysis.

[10] PRAMS reports income in brackets, which we include as categorical controls.

hypertension or diabetes,[11] and urban versus rural residence. State-by-year controls account for time-varying economic and healthcare characteristics, including the fertility rate (CDC Vital Statistics), Medicaid managed care penetration (Centers for Medicare & Medicaid Services), Medicaid eligibility thresholds for pregnant adults (KFF), federally qualified health centers per capita (Health Resources and Services Administration), unemployment rate (Bureau of Labor Statistics), an indicator for the Affordable Care Act's Medicaid expansion (KFF), and number of dentists per capita (Area Health Resources File). Standard errors are clustered at the state level and estimates incorporate PRAMS survey weights at the individual level to ensure population representativeness. We report both cluster-robust standard errors and wild cluster bootstrap p-values using Webb weights given the limited number of state clusters (N=13) (Cameron et al., 2008; Webb, 2023). Cluster-robust standard errors are likely to overstate statistical significance in our setting whereas the wild cluster bootstrap is generally substantially more conservative .[12] We examine heterogeneity in policy impacts across demographic subgroups by estimating stratified versions of Equation (1) by age, race and ethnicity, marital status, and education.

We also estimate event study models to evaluate outcome pre-trends and the evolution of post-policy implementation effects. These models are specified as follows:

$$Y_{istd} = \alpha_0 + \sum_{\tau=-\kappa_a}^{-2} \alpha_\tau \cdot 1(TSE_{std} = \tau) + \sum_{\sigma=0}^{\kappa_b} \delta_\sigma \cdot 1(TSE_{std} = \sigma) + \mathbf{X}_{istd}\boldsymbol{\beta} + \mu_{sd} + \lambda_t + \epsilon_{istd} \quad (2)$$

where TSE_{std} denotes the number of months since policy implementation in state s within sub-experiment d, with α_τ and δ_σ capturing pre-policy and post-policy event-time effects, respectively. All other variables are defined as before. Our event study specification aligns birth year and month with the timing of Medicaid dental benefit implementation in each state and groups the pre- and post-treatment periods into three-month intervals. We include eight pre- and post-policy implementation indicators (i.e., covering up to 22 months or more both preceding and following policy changes). The three-month period directly preceding any exposure to Medicaid dental benefits serves as the reference category. This structure allows for a detailed examination of how outcomes evolve over time. The model differentiates between partial policy exposure, observed in the initial post-policy intervals, and full policy exposure, captured in later periods. Given typical gestation lengths, δ_4 and subsequent post-policy indicators capture

[11] We include this control to proxy for a mother's health status before conception

[12] We also explored clustering standard errors at the state-by-sub-experiment level, however this requires an assumption of independence across clusters which is unlikely to be met in our setting as all sub-experiments contain the same control states. These results were uniformly less conservative than inference based on both the cluster-robust and wild cluster bootstrap methods with clustering at the state level shown in the paper. Results are available upon request.

full policy exposure, with exposure increasing across δ_1 through δ_3. By aligning events temporally within states, this approach provides a clearer understanding of the timing, stabilization, and persistence of policy impacts, offering insights into both short-term effects and longer-term trends.

Equation (2) tests the plausibility of the parallel trends assumption, which is critical to a causal interpretation of our estimates. Significant coefficient estimates for the pre-period(α_τ), or estimates that exhibit an apparent increasing or decreasing trend, could indicate bias. The post-policy coefficients δ_σ capture the dynamic effects of the policy, and in our context, whether impacts differ according to partial vs. full exposure during pregnancy. We present event study estimates with 95% confidence intervals based on cluster-robust standard errors, which biases us toward estimating significant pre-period effects.

For our supplementary analyses using BRFSS, we use a difference-in-differences framework adapted to the structure of the BRFSS data.[13] We restrict the sample to reproductive-aged women (ages 18-44) with household incomes below the Medicaid income eligibility limit in their state of residence during the year of interview. We estimate models that replace the pregnancy dental benefit indicator with an indicator for Medicaid dental benefits for non-pregnant adults.

In an extension to our main analysis, we explore the role of pre-conception Medicaid dental benefits two ways. First, we restrict our analysis to control states and the three treatment states that adopted pregnancy and non-pregnancy dental benefits simultaneously (CO, IL, and SC), excluding the two treatment states that adopted pregnancy dental benefits in isolation (UT and VA). We then estimate Equation (1) on this sample and two income-based sub-samples: women with incomes above and below 138% of the federal poverty level (FPL). Women with incomes above 138% FPL would be unlikely to qualify for pre-conception Medicaid since this was the highest eligibility limit for non-pregnant adults among the treatment states during most of the pre-adoption period.[14] Conversely, women with incomes up to 138% FPL may have qualified for pre-conception Medicaid and therefore may have had access to pre-conception Medicaid dental benefits.[15] Second, we augment Equation (1) by including a control for exposure to

[13]This analysis examines a different policy - non-pregnancy dental benefits - with different treatment variation including more frequent dropping of dental benefits and reversible treatment within states. Thus, we use a two-way-fixed effects specification and also include all available years of data from 2000-2022 in these analyses.

[14]Illinois' income eligibility limit for parents was 191% FPL in January 2012 and subsequently dropped to 138% FPL in 2013 and later years. Colorado's income eligibility limit for parents was 106% in January 2012 and increased to 138% FPL in January 2014. South Carolina's income eligibility limit for parents was 91% in January 2012 and decreased to 67% in January 2014 and subsequent years. These data are available from KFF: https://www.kff.org/medicaid/state-indicator/medicaid-income-eligibility-limits-for-parents/

[15]We use reported household income during pregnancy, which may differ from pre-pregnancy income levels, but is likely to be correlated.

non-pregnancy dental benefits in the one year prior to imputed conception date,[16] while including the full analysis sample. Taken together, these approaches allow us to assess the role of pre-conception access to Medicaid dental benefits and whether the estimated effects of pregnancy dental benefits persist when accounting for these potential impacts,

We conduct several additional robustness checks to assess the stability of our results. First, we extend our analysis period through 2021 and re-estimate our models including two additional policy changes (Delaware and West Virginia). For these extended analyses, we add state-level and pregnancy-period COVID-19 controls—including case and death rates, business closure stringency, the Oxford Government Response Index, and indicators for stay-at-home and elective-procedure restrictions—to account for pandemic-related disruptions in care. Second, we vary the length of the event window to 24, 36, and 60 months to examine sensitivity to the choice of pre- and post-policy observation periods in the stacked DiD framework. Third, we assess robustness to alternative definitions of Medicaid enrollment by varying the use of birth certificate payer information and self-reported Medicaid coverage.

Fourth, we consider conception-based exposure measures as an alternative to our main analysis that uses birth date to assign treatment status. While this analysis allows us to account for the duration of exposure, it also requires imputation as we do not have access to information on the exact day of birth or exact gestational timing. After imputing conception date based on month and year of birth and reported weeks of gestation,[17] we then assign each pregnancy a standardized 40-week gestation over which policy exposure may occur. This avoids endogenously assigning shorter exposure to preterm and shorter gestation pregnancies. We replicate our main analysis by examining any exposure to pregnancy dental benefits based on this definition and also examine an indicator for exposure by the first trimester and a continuous measure for the percent of exposure during pregnancy. This analysis replaces birth quarter-by-year fixed effects with conception quarter-by-year fixed effects.[18]

Fifth, we conduct "leave-one-out" analyses, removing each treatment state from the analysis in turn, to explore the contributions of each treatment state to our findings. Sixth, we replicate our main analyses, except we estimate two-way-fixed effects models for comparison with our main stacked DiD analysis. Finally, while our primary analysis focuses on individuals enrolled in Medicaid during pregnancy—those directly exposed to

[16]Since we do not observe exact day of birth, we impute conception date by assuming that births occur in the middle of the month, and then subtract gestation in weeks.

[17]Data for Virginia in 2014 is excluded from these analyses due to missing gestational length information required to construct conception-based exposure measures. We include Virginia's data for 2012-2013 and 2015-2019.

[18]In our main analysis, policy exposure is exogenous conditional on birth timing - e.g., births occurring in the same month and year are equally likely to be considered as exposed regardless of the pregnancy/infant health outcome. Similarly, in our conception date analysis, exposure is exogenous conditional on conception date timing.

Medicaid pregnancy dental benefits—we replicate our analyses for alternative samples, including all women, women with less than a bachelor's degree and those with a bachelor's degree or higher, and women with household incomes up to $50,000 and with incomes of $75,000 or more. Results for all women and lower-education or lower-income subgroups provide intent-to-treat estimates, mitigating concerns about selection into Medicaid when pregnancy dental benefits are offered, while estimates for higher-education and higher-income subgroups are expected to be smaller because women in these groups are less likely to be Medicaid eligible.

5 Results

5.1 Effects of Medicaid Pregnancy Dental Benefits on Dental Cleanings

Table 3 presents the results of our main specification shown in Equation (1) . We estimate that Medicaid pregnancy dental benefits increase the probability of having received a dental cleaning during pregnancy by 9.5 percentage points (Table 3, Column (1)). Given the baseline mean of 25.5%, this estimate represents a 37% relative increase. Results for dental cleanings are similar across specifications with and without individual and state-by-year controls, ranging from 8.2 (32.2% increase) to 9.5 (37.2% increase) percentage points (Table A.2). Statistical significance is robust to inference based on wild cluster bootstrap p-values, addressing concerns related to the limited number of state-level treatment clusters.

(Table 3 here)

Figure 2 provides event study estimates of the impact of Medicaid pregnancy dental benefits on prenatal dental cleaning rates from Equation (2). Before policy implementation, most estimates are relatively small in magnitude and statistically insignificant with no clear trend. After policy implementation, all estimates are positive and statistically significant, with a slight increasing trend over time indicating that impacts may grow modestly with additional exposure. By the later post-policy months, the effect plateaus at a consistently higher rate of dental cleaning compared to pre-policy levels, indicating a lasting shift in healthcare-seeking behavior among Medicaid-enrolled pregnant individuals.

(Figure 2 here)

We examine heterogeneity in policy impacts across demographic subgroups by estimating stratified versions of Equation (1) by age, race and ethnicity, marital status,

and education. We find broadly consistent increases in prenatal dental cleaning following Medicaid pregnancy dental benefit expansions (Figure 3). Estimated effects were somewhat larger among non-Hispanic Black individuals, married individuals, and those with higher education levels. However, 95% confidence intervals overlapped across these analyses, with point estimates suggestive of larger impacts among married women and those with more education. We note, however, that we may have lacked power to detect significant differences for some smaller subgroups including Hispanic, non-Hispanic Black, and ages 35 and older.

(Figure 3 here)

5.2 Effects of Medicaid Pregnancy Dental Benefits on Birth Outcomes

Table 3 (Columns (2) to (5)) presents our main findings for birth outcomes from Equation (1). Medicaid pregnancy dental benefits significantly reduce the likelihood of preterm birth by 1.35 percentage points, representing an 11.5% decrease relative to the baseline mean of 11.7%. Additionally, we observe statistically significant reductions in very low birth weight by 0.47 percentage points, representing a 26% decline relative to the baseline mean (1.8%), and in low birth weight by 0.59 percentage points, representing a 5.5% decline relative to the baseline mean (10.7%). While all three effects are statistically significant using conventional cluster-robust standard errors, inference based on wild cluster bootstrap p-values indicates that the reduction in preterm birth remains statistically significant at conventional levels, whereas the reductions in very low birth weight and low birth weight are marginally significant at the 10% level. We additionally estimate a reduction in small for gestational age by 2.19 percentage points (an 18% reduction relative to a mean of 12.2%), which is statistically significant under conventional clustered inference but becomes imprecise under wild cluster bootstrap inference ($p = 0.23$).

Figure 4 presents event study estimates for each birth outcome from Equation (2). Prior to policy implementation, coefficients for most outcomes fluctuate around zero with overlapping confidence intervals, supporting the plausibility of parallel trends between treated and control states. An exception is low birth weight, for which pre-policy estimates deviate from zero; however, these estimates are uniformly negative and exhibit a relatively flat pattern that mirrors the post-policy period. Following policy implementation, we observe negative effects for small for gestational age, preterm birth, and very low birth weight, while there is no clear or systematic evidence of an effect on low birth weight. We therefore de-emphasize findings for low birth weight in our subsequent discussion.

(Figure 4 here)

We assess potential heterogeneity in the effects of Medicaid pregnancy dental benefit expansions on birth outcomes across demographic subgroups. These estimates are somewhat noisy, particularly for the lower-prevalence outcomes. Focusing on preterm birth, very low birth weight, and small for gestational age where we found evidence of declines that did not appear to violate parallel trends, 95% confidence intervals generally overlapped across subgroups. Nonetheless, estimates suggested larger declines in small for gestational age among married women (Figure A.1), which was consistent with estimates for dental cleanings.

5.3 Extensions and Mechanisms

Table 4 reports estimates of the effect of Medicaid pregnancy dental benefits on prenatal dental cleaning and birth outcomes after restricting the sample to three treatment states that adopted both pregnancy and non-pregnancy dental benefits (CO, IL, SC) and control states . This analysis allows us to assess whether exposure to dental coverage prior to pregnancy plays a role in dental care utilization during pregnancy and subsequent birth outcomes. The direction of this effect is theoretically ambiguous. Women who recently received dental care before conception may be less likely to seek additional care during pregnancy due to reduced immediate need. Conversely, women who have established relationships with dental providers or received counseling on recommended oral health practices may be more likely to continue preventive dental visits once pregnant. Access to pre-conception dental benefits may therefore either amplify or depress the impacts of pregnancy dental benefits on dental care utilization and infant health outcomes.

Across all income levels (Panel A), as shown in Table 4, we estimate that pregnancy dental benefits significantly increase prenatal dental visits by 10 percentage points, which is similar to the corresponding estimate in Table 3. We also observe economically meaningful improvements in birth outcomes. The likelihood of being small for gestational age declines by 2.88 percentage points, the probability of preterm birth falls by 1.39 percentage points, and very low birth weight falls by 0.58 percentage points.[19] While all effects are statistically significant using conventional clustered standard errors, wild cluster bootstrap inference indicates that only the reduction in preterm birth is marginally significant ($p = 0.080$), while the reductions in small for gestational age and very low birth weight become imprecise under wild cluster bootstrap inference ($p = 0.204$ and $p = 0.218$, respectively). These effects are relatively similar to those estimated in the main analysis.

(Table 4 here)

[19]We also observe a 0.53 percentage point reduction in low birth weight that is only significant at the 10% level. However, our analysis of pretrends suggests that this result may be unreliable.

Panels B and C in Table 4 stratify the sample by household income relative to 138% FPL, the threshold for adult Medicaid eligibility under the Affordable Care Act. This stratification distinguishes women more versus less likely to have had access to adult Medicaid dental coverage prior to conception and allows us to assess whether the estimated effects of pregnancy dental benefits persist among women likely experiencing Medicaid dental coverage for the first time during pregnancy. Among women with household incomes at or below 138% FPL, who are more likely to have been eligible for adult Medicaid dental coverage prior to conception, Medicaid pregnancy dental benefits are associated with a 9.5 percentage point increase in prenatal dental cleaning. We also observe reductions in adverse birth outcomes, including a 1.65 percentage point decline in small for gestational age, a 1.61 percentage point decline in preterm birth, and a 0.92 percentage point reduction in very low birth weight.[20] While all effects except preterm birth are statistically significant using conventional clustered standard errors, inference based on wild cluster bootstrap p-values indicates that only the increase in dental cleaning remains statistically significant ($p = 0.019$), which is likely due to reduced power given fewer treatment states and smaller sample sizes in this analysis.

Among women with household incomes above 138 percent of the federal poverty level, who are less likely to have had access to adult Medicaid dental coverage prior to conception, Medicaid pregnancy dental benefits are associated with a larger 13.3 percentage point increase in prenatal dental cleaning. We also estimate a 4.7 percentage point reduction in small for gestational age. While both effects are statistically significant using conventional clustered standard errors, wild cluster bootstrap inference indicates that the increase in dental cleaning is marginally significant ($p = 0.058$), while the reduction in small for gestational age becomes imprecise ($p = 0.146$). While not statistically significant, we also estimate a 0.97 percentage point reduction in preterm birth. Estimated effects on other birth outcomes are small in magnitude and not statistically significant. Nonetheless, while these results are less precise, they are consistent with impacts on birth outcomes among women unlikely to have exposure to pre-conception Medicaid dental benefits.[21] While not statistically distinguishable, they are also suggestive of a stronger impact on dental care utilization among women newly gaining access to dental coverage during pregnancy relative to those that had access both before and during pregnancy.

We also replicate our main analysis with the addition of a control for exposure to pre-conception Medicaid dental benefits in the year prior to the imputed conception date (Table 5). Pregnancy dental benefits continue to significantly increase prenatal dental

[20] We also observe a 0.91 decline in low birth weight, however given evidence of possible preexisting outcome trends, this result may be unreliable.

[21] This somewhat higher income sample of women with pregnancy Medicaid coverage may also experience different impacts on birth outcomes for other reasons, such as differences in pre-pregnancy private dental insurance and baseline health.

visits by 8.8 percentage points, whereas pre-conception dental benefits have small and statistically insignificant effects. For birth outcomes, while pregnancy dental benefits are associated with statistically significant reductions in small for gestational age and very low birth weight using conventional clustered standard errors, wild cluster bootstrap inference indicates that these effects become imprecise for small for gestational age (p = 0.319), while the reduction in very low birth weight is marginally significant (p = 0.097). Pre-conception dental benefits are generally not significantly associated with the birth outcomes considered except for an estimated 2.10 percentage point decline in preterm birth that is only marginally significant with cluster-robust standard errors. Taken together, the findings point to a meaningful role for pregnancy-specific dental coverage in increasing access to preventive dental care during pregnancy and potentially improving early-life health outcomes.

(Table 5 here)

Table 6 examines potential mechanisms through which Medicaid pregnancy dental benefits may influence birth outcomes. We find that policy adoption is associated with sizable increases in the likelihood of having dental insurance coverage (13.5 percentage points) and receiving advice from a provider on how to care for teeth and gums (5.8 percentage points). Inference based on wild cluster bootstrap p-values indicates that the increase in dental insurance coverage remains statistically significant (p = 0.045), while the increase in receipt of dental care advice is marginally significant (p = 0.07). These findings indicate that women are aware of Medicaid dental insurance and suggest a likely decline in out-of-pocket costs for dental services, supporting the observed increase in dental visits. Moreover, receipt of counseling may improve oral healthcare behaviors at home, complementing in-office care in improving oral health. However, while the point estimate is positive (1.1 percentage points), we do not estimate a statistically significant change in reporting that oral health during pregnancy is important under either conventional clustered inference or wild cluster bootstrap inference.

When examining broader maternal health behaviors, we find no evidence that Medicaid pregnancy dental benefits affect the likelihood of receiving adequate prenatal care, as measured by the Kotelchuk Index. We estimate declines in any maternal smoking during pregnancy (1.49 percentage point decrease) and smoking during the last three months of pregnancy (1.21 percentage point decrease) and an increase in reporting non-smoking status following pregnancy (1.24 percentage point increase). However, these estimates are generally statistically indistinguishable from zero, with the exception of a marginally significant effect on any smoking during pregnancy when using cluster-robust inference only.

(Table 6 here)

We turn to the BRFSS to assess possible improvements in systemic health that could improve the fetal environment. We estimate that adult Medicaid dental coverage is associated with a 1.3 percentage point increase in the probability of reporting that general health status is good, very good, or excellent (vs. fair or poor) among Medicaid income-eligible reproductive-age women, and a 1.2 percentage point increase in the probability of reporting excellent health status relative to all other categories (Table A.3). Given the absence of a comparable general health measure in PRAMS, this complementary analysis supports the broader argument that increased access to dental care under Medicaid may enhance overall well-being among low-income women.

5.4 Robustness Checks

To evaluate the robustness of our findings, we conduct a range of supplementary analyses presented in the appendix. While our main analysis excludes data for 2020 and 2021, event study estimates that include the pandemic period continue to display patterns consistent with our main results (Figures A.2, A.3 and Table A.4). Event window sensitivity checks indicate that estimates are relatively stable across windows, though precision varies and birth outcome estimates are attenuated in some instances (Table A.5). Estimates are also relatively robust across different samples that use alternative definitions of Medicaid enrollment (Table A.6). In our analysis using conception-based exposure measures, estimates for prenatal dental cleaning remain robust, and effects on small for gestational age are consistently negative and significant when using conventional cluster-robust inference, though not when considering wild cluster bootstrap inference (Table A.7). Results for other birth outcomes are generally attenuated. Leave-one-out tests confirm that no single treated state drives the results (Table A.8). Estimates for dental cleanings are very similar to our main results when we use a two-way fixed effects specification (Table A.9). Point estimates for the birth outcomes remain negative and are of a similar magnitude as in our main results, but are generally less precise. Finally, in alternative intent-to-treat samples based on education and income, we estimate statistically significant increases in dental cleanings for most groups, except for those with incomes of at least $75,000 (Table A.10). As expected, estimates are smaller in magnitude than when we restrict to Medicaid enrollees as in our main analysis. Furthermore, point estimates are larger among groups more likely to be Medicaid eligible (i.e., lower education and lower income). For birth outcomes, estimates are generally less precise; however, we do observe a statistically significant reduction in very low birth weight among women with lower educational attainment, consistent with the pattern of larger effects in populations more likely to be affected by Medicaid pregnancy dental benefits.

6 Conclusion

In this paper, we find robust evidence that Medicaid pregnancy dental benefits increase access to preventive dental care during pregnancy. Our main estimate (9.5 percentage point increase) is much smaller than that from research using the PRAMS to examine Virginia's addition of a comprehensive dental benefit (17 percentage point increase) (Naavaal & Harless, 2022). Moreover, we find significant increases in dental cleanings within the first year of policy implementation whereas the previous analysis indicated that effects only became significant four years after Virginia's 2015 policy change. While both papers use the PRAMS, our analyses depart in several ways such as the inclusion of control states, multiple treatment states, and differences in controls for time trends and time-varying state-level factors.

By contrast, our estimates align with similar analyses of Medicaid dental benefits among non-pregnant adults, which indicate increases in annual dental visit rates ranging from 6-14 percentage points (Decker & Lipton, 2015; Abdus & Decker, 2019; Lipton, 2021). It is an empirical question whether Medicaid dental benefits are likely to have larger impacts on pregnant women compared with other adults. On one hand, effects may be smaller among pregnant individuals because coverage applies for a short duration, limiting the time available to seek care, and because some pregnant patients and providers remain cautious about receiving or delivering dental procedures during pregnancy. On the other hand, pregnancy represents a period of heightened motivation and frequent contact with the healthcare system, which could amplify responsiveness once financial barriers are removed. It is also worth noting that a substantial proportion of policy variation in previous research included coverage reductions (Decker & Lipton, 2015; Abdus & Decker, 2019; Lipton, 2021) whereas our analysis of pregnancy dental benefits only included coverage adoptions, which could influence comparisons if adding and dropping benefits have asymmetric effects .

We also find suggestive evidence that pregnancy dental benefits may reduce adverse birth outcomes including preterm birth and very low birth weight. Our main estimates—an 11.5% reduction in preterm births and a 26% decline in very low birth weight—are relatively large in magnitude, but consistent with the upper range of effects documented in related clinical and observational studies. While clinical evidence is mixed, periodontal intervention during pregnancy has been found to reduce preterm birth by 15-25% in some studies (Iheozor-Ejiofor et al., 2017; Puertas et al., 2018; Castaño-Suárez et al., 2024). Moreover, one analysis of linked medical and dental claims found that periodontal intervention preceded a 74% reduction in spending on atypical pregnancy care (Jeffcoat et al., 2014), again suggesting the potential for relatively large-scale impacts.

Parallel evidence from quasi-experimental studies of Medicaid eligibility expansions demonstrates that increasing coverage during pregnancy may improve prenatal care access

and can lead to measurable improvements in infant health, though evidence is mixed (Currie & Gruber, 1996; Howell, 2001; Sonchak, 2015; Saygili & Bayindir, 2024). In terms of research indicating gains, Currie & Gruber (1996) find that Medicaid eligibility expansions during the 1980s reduced infant mortality by roughly 8%, while Howell (2001) document that state-level expansions of pregnancy-related Medicaid eligibility in the 1990s increased prenatal care use and reduced the likelihood of low birth weight among low-income women. More recent evidence from a border-county difference-in-differences study of Arkansas's ACA Medicaid expansion shows a 1.4-percentage-point decline in preterm birth (about a 20% relative reduction) and a 1.8-percentage-point decline in low birth weight (about a 40% relative reduction) among White infants relative to neighboring Texas (Saygili & Bayindir, 2024). Overall, our estimates fall within this documented range of impacts.

In our analysis of mechanisms, we find increases in dental insurance coverage and provider dental care advice, both of which are likely to facilitate earlier or more consistent dental care during pregnancy. We do not find strong evidence that the observed effects are driven by significant changes in prenatal care adequacy or maternal smoking behavior, though point estimates for smoking during pregnancy are negative. Instead, the pattern of findings points to direct oral-systemic health mechanisms—such as reductions in periodontal inflammation and systemic bacterial load—which have been documented as risk factors for adverse pregnancy outcomes. We find some support for the notion that access to dental care can improve systemic health among a similar population of reproductive-age women in our analysis of the BRFSS where we estimate that Medicaid dental benefits are associated with a modest but significant improvement in general reported health status. This finding is also consistent with established linkages between improved oral health and systemic health, including inflammation and chronic disease risk (Oliveira et al., 2010; Chee et al., 2013; Manger et al., 2017; Sanz et al., 2020).

These findings should be viewed within the broader policy environment in which Medicaid operates. Recent federal legislation and budgetary adjustments—including changes to Medicaid financing—underscore how state-level benefits can be affected by broader economic conditions and federal policy decisions. In times of budget tightening, optional services like dental coverage may be among the first to face cuts or reduced support.

Overall, our findings provide evidence that pregnancy dental benefits induce meaningful increases in dental care use among a population with a high prevalence of oral health problems and a low rate of dental visits. We also find suggestive evidence that dental benefits may improve birth outcomes. While most states currently provide some form of a pregnancy dental benefit, mounting pressure on state Medicaid budgets may prompt reconsideration of optional coverage. As policymakers consider future

modifications to Medicaid, this research could inform decisions on whether or not to maintain or expand coverage of dental services for pregnant populations.

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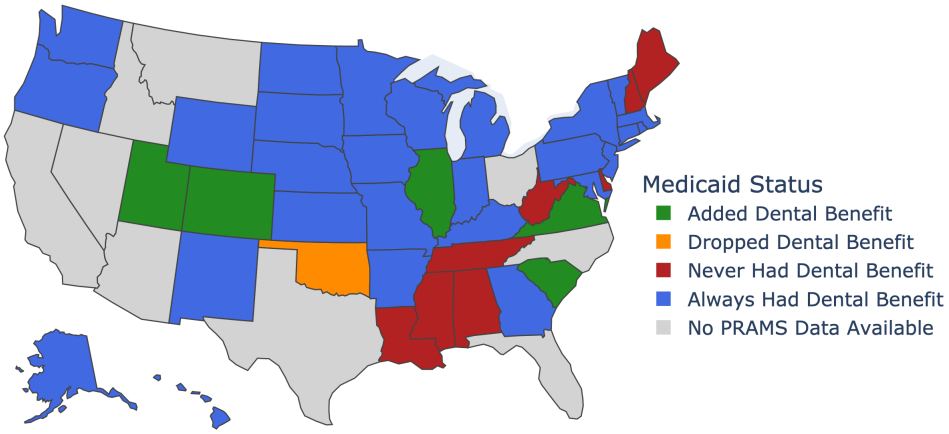
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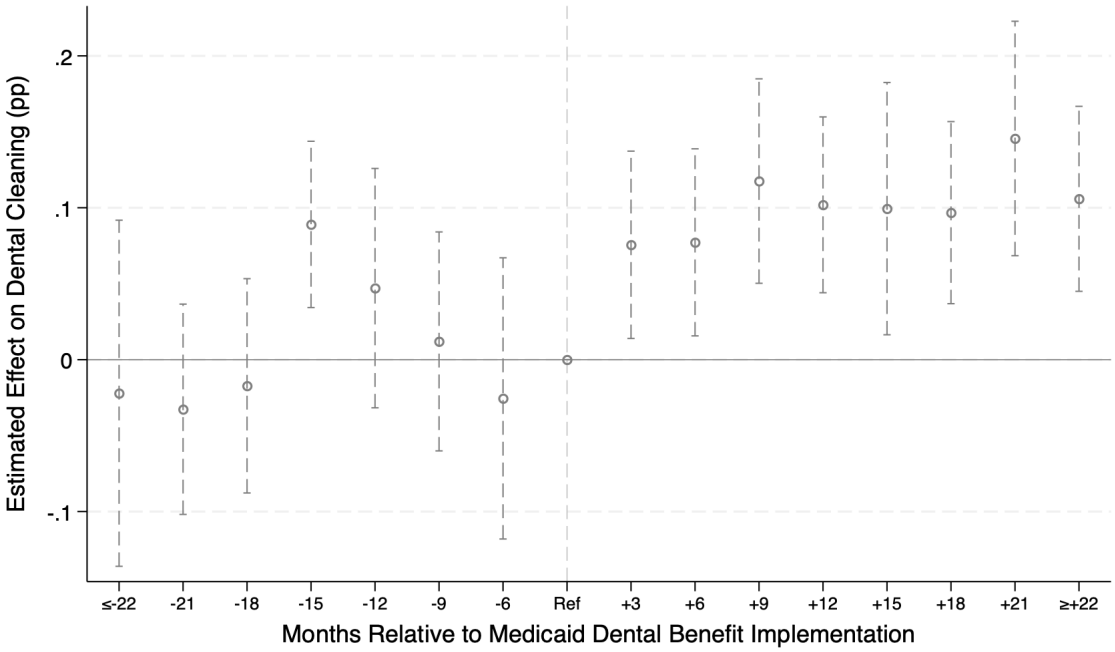
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Figure 1: State Variation in Medicaid Pregnancy Dental Benefits, 2012-2019



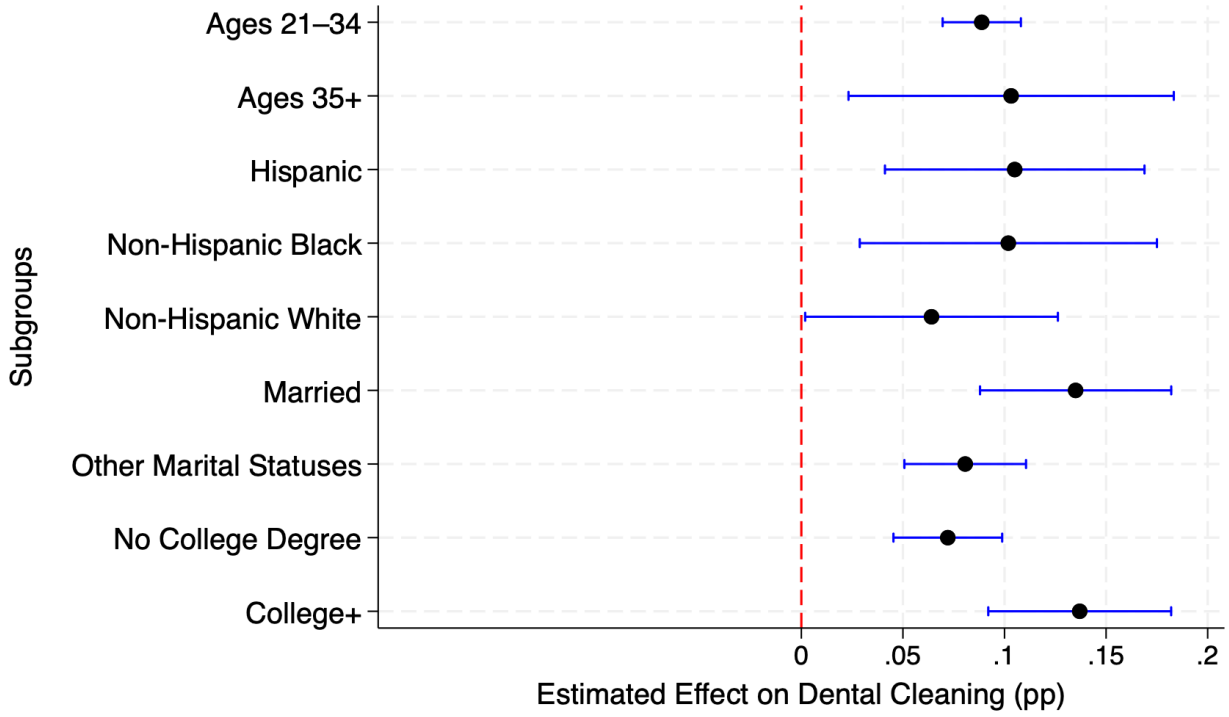
Notes: Figure shows state-level variation in Medicaid pregnancy dental benefit status from 2012-2019, based on the Pregnancy Risk Assessment Monitoring System (PRAMS) and state Medicaid policy reports. States shown in green added pregnancy dental benefits during the study period, blue states maintained pre-existing benefits, and red states did not offer pregnancy dental benefits. States shown in gray did not contribute sufficient PRAMS data or fell below reporting thresholds for inclusion during the study period.

Figure 2: Event Study Estimates of the Effect of Medicaid Pregnancy Dental Benefits on Prenatal Dental Cleanings, 2012-2019 PRAMS



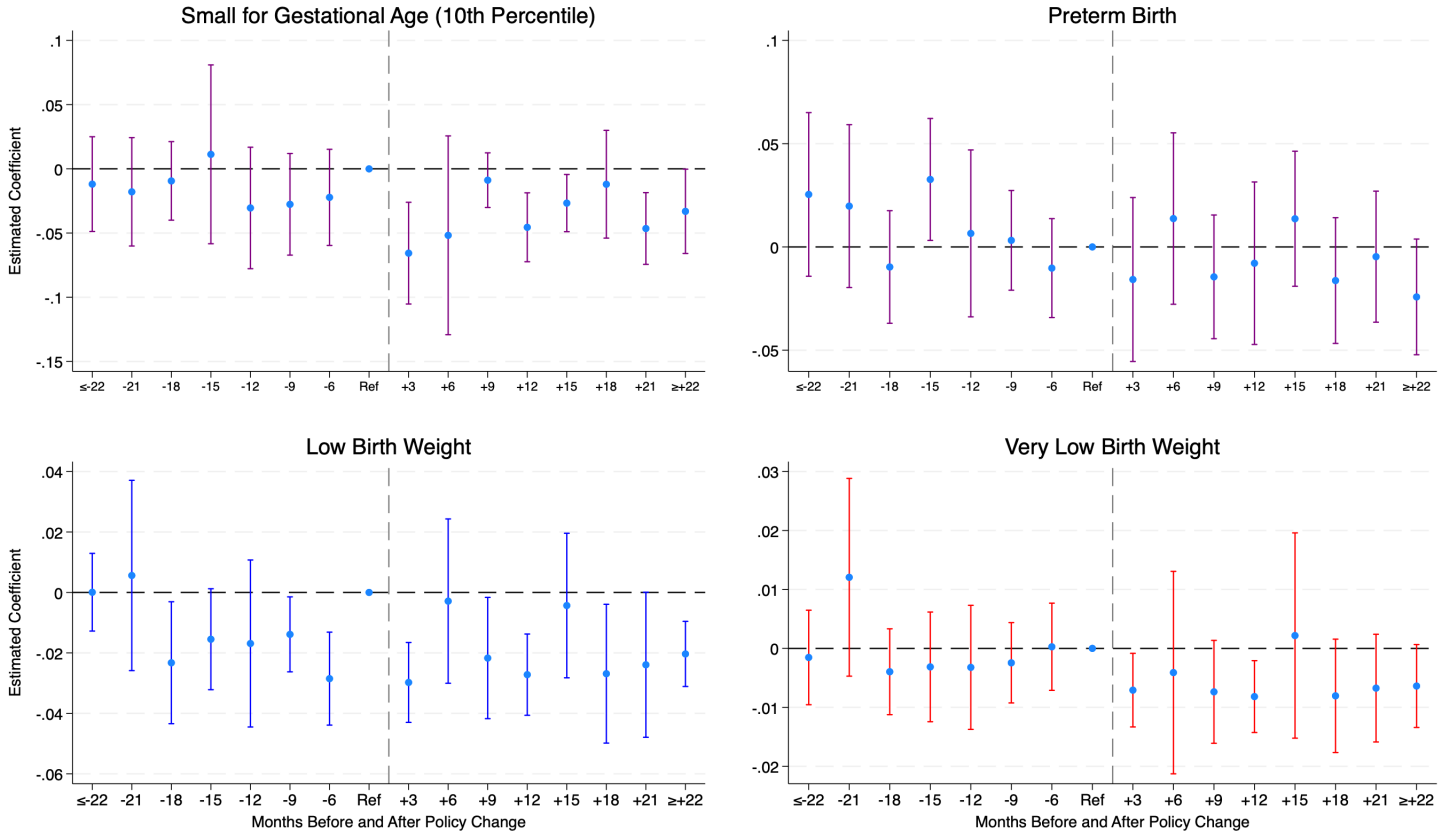
Notes: Data come from the Pregnancy Risk Assessment Monitoring System (PRAMS), 2012-2019. The figure displays event study estimates of the effect of Medicaid pregnancy dental benefit expansions on prenatal dental cleaning rates, using a stacked difference-in-differences approach, as described in the text. The x-axis represents time in months before and after policy adoption, and the y-axis shows the estimated coefficients. Cluster-robust 95% confidence intervals are indicated by the vertical lines around the point estimates. The reference group is set to 0-3 months before policy implementation, and the estimates test the parallel trends assumption while capturing dynamic treatment effects.

Figure 3: Heterogeneous Effects of Medicaid Pregnancy Dental Benefits on Prenatal Dental Cleanings by Subgroup, 2012-2019 PRAMS



Notes: Data come from the Pregnancy Risk Assessment Monitoring System (PRAMS), 2012-2019. Figure presents heterogeneous effects of Medicaid pregnancy dental benefits on prenatal dental cleaning rates across demographic subgroups: maternal age (21-34, 35+), race/ethnicity (Hispanic, Non-Hispanic Black, Non-Hispanic White), marital status (married vs. other), and education (no college degree vs. college+). Each point shows the estimated coefficient from a subgroup-specific stacked difference-in-differences regression, with horizontal bars representing cluster-robust 95% confidence intervals. The vertical red line at zero represents no effect.

Figure 4: Event Study Estimates of the Effect of Medicaid Pregnancy Dental Benefits on Birth Outcomes, 2012-2019 PRAMS



Notes: Data come from the Pregnancy Risk Assessment Monitoring System (PRAMS), 2012-2019. The figure displays event study estimates of the effect of Medicaid pregnancy dental benefit expansions on key birth outcomes, including small for gestational age (10th percentile), preterm birth, low birth weight, and very low birth weight, using a stacked difference-in-differences approach. Each panel corresponds to a separate outcome variable. The x-axis represents time in months before and after policy adoption, and the y-axis shows the estimated coefficients. Cluster-robust 95% confidence intervals are indicated by the vertical lines around the point estimates. The reference group is set to 0-3 months before policy implementation, and the estimates test the parallel trends assumption while capturing dynamic treatment effects.

Table 1: Summary Statistics by State Medicaid Pregnancy Dental Benefit Adoption Status, 2012-2019 PRAMS

	(1)	(2)	(3)
	No Dental Benefit	**Added Dental Benefit**	
		Pre	Post
Dental Care Utilization			
Received Dental Cleaning	0.251	0.271	0.304
Birth Outcomes			
Preterm Birth	0.117	0.107	0.093
Small for Gestational Age (10th Percentile)	0.122	0.112	0.11
Low Birth Weight	0.108	0.091	0.091
Very Low Birth Weight	0.017	0.016	0.016
Demographics			
Age of Mother	27.34	27.63	28.12
Less than High School	0.202	0.211	0.198
High School Graduate (GED)	0.399	0.345	0.356
Some College or Associate Degree	0.319	0.341	0.339
Bachelor's Degree or Higher	0.079	0.103	0.107
Non-Hispanic White	0.476	0.413	0.378
Non-Hispanic Black	0.337	0.261	0.246
Hispanic	0.11	0.281	0.286
Non-Hispanic Other	0.064	0.019	0.057
Married	0.342	0.402	0.418
Urban Areas	0.717	0.692	0.787
Household Income (Midpoint)	19776.96	18738.94	22149.05
Pre-pregnancy Risk	0.077	0.072	0.072
State Controls			
Medicaid Eligibility Limit for Pregnant Women	2.12	1.99	2.08
ACA Medicaid Expansion	0.054	0	0.405
Dentists per Capita	47.85	61.12	63.62
Unemployment Rate	7	8.76	5.89
Fertility Rate	62.29	63.37	62.29
Managed Care Organization	60.77	25.65	42.72
Health Centers per Capita	4.64	3.49	3.53

Notes: Data come from the Pregnancy Risk Assessment Monitoring System (PRAMS), 2012-2019. Table reports survey-weighted means of maternal, demographic, and state-level characteristics by Medicaid pregnancy dental benefit status and period. "No Dental Benefit" refers to control states that did not adopt pregnancy dental benefits during the study period. "Pre" and "Post" refer to births occurring before and after the effective date of benefit implementation in treatment states.

Table 2: Medicaid Pregnancy Dental Benefit Expansion Sub-experiments

(1) **Sub-experiment Date**	(2) **Event Window**	(3) **Treatment States**	(4) **Control States**
10/2013	01/2012–10/2017	UT	AL, DE, LA, ME, MS, NH, TN, WV
04/2014	01/2012–04/2018	CO	AL, DE, LA, ME, MS, NH, TN, WV
07/2014	01/2012–07/2018	IL	AL, DE, LA, ME, MS, NH, TN, WV
12/2014	01/2012–12/2018	SC	AL, DE, LA, ME, MS, NH, TN, WV
03/2015	01/2012–12/2018	VA	AL, DE, LA, ME, MS, NH, TN, WV

Notes: This table summarizes the sub-experiments used in the stacked difference-in-differences design. Each sub-experiment corresponds to a treatment state's Medicaid pregnancy dental benefit adoption date and compares it to a set of control states that did not adopt similar benefits during the event window. Event windows span up to 48 months before and after policy implementation.

Table 3: Stacked DiD Estimates of the Effect of Medicaid Pregnancy Dental Benefits on Dental Cleaning and Birth Outcomes, 2012-2019 PRAMS

	(1) Dental Cleaning	(2) Small for Gestational Age (P10)	(3) Preterm Birth	(4) Low Birth Weight	(5) Very Low Birth Weight
Pregnancy Dental Benefit	0.0949***	-0.0219***	-0.0135***	-0.00590*	-0.00473***
	(0.0105)	(0.00568)	(0.00399)	(0.00302)	(0.00109)
Baseline Mean	0.255	0.122	0.117	0.107	0.018
Wild cluster bootstrap p-value	0.002	0.230	0.036	0.064	0.079
Observations	94612	86525	90604	95130	95130
R-squared	0.035	0.013	0.024	0.022	0.009

Notes: Data come from the Pregnancy Risk Assessment Monitoring System (PRAMS), 2012-2019. This table reports estimates of the effect of Medicaid pregnancy dental benefits on prenatal dental cleaning (Column (1)) and birth outcomes (Columns (2)-(5)), as described in the text. Each column represents results from a separate regression estimated using a stacked difference-in-differences framework that accounts for staggered policy adoption. Covariates include maternal age, race/ethnicity, education, marital status, household income, urban/rural residence, and pre-pregnancy risk factors, and state-by-year controls including Medicaid eligibility limits, managed care penetration, and broader socioeconomic indicators. All models include state-by-sub-experiment and birth quarter-by-year fixed effects and incorporate PRAMS survey weights to ensure population representativeness. Cluster-robust standard errors clustered at the state level are reported in parentheses, and statistical significance stars are based on these standard errors (*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$). Wild cluster bootstrap p-values clustered at the state level using Webb weights are reported in a separate row to provide inference robust to the limited number of state clusters.

Table 4: Stacked DiD Estimates of the Effect of Medicaid Pregnancy Dental Benefits on Dental Cleaning and Birth Outcomes, Excluding Pregnancy-Only Dental Coverage States, 2012-2019 PRAMS

	(1) Dental Cleaning	(2) Small for Gestational Age (P10)	(3) Preterm Birth	(4) Low Birth Weight	(5) Very Low Birth Weight
Panel A. All Income Levels					
Pregnancy Dental Benefit	0.0352***	-0.0113	-0.0161***	-0.00175	0.000857
	(0.0108)	(0.00735)	(0.00525)	(0.00343)	(0.000915)
Baseline mean	0.257	0.121	0.117	0.108	0.017
Wild cluster bootstrap p-value	0.039	0.340	0.216	0.705	0.553
Observations	57532	52566	55085	57884	57884
R-squared	0.018	0.005	0.007	0.004	0.002
Panel B. Low Income (≤138% FPL)					
Pregnancy Dental Benefit	0.0255	-0.0268***	-0.00777	-0.00327	-0.00113
	(0.0160)	(0.00634)	(0.00838)	(0.00419)	(0.00237)
Baseline mean	0.232	0.127	0.124	0.117	0.019
Wild cluster bootstrap p-value	0.154	0.163	0.503	0.412	0.732
Observations	35811	32930	34435	36030	36030
R-squared	0.024	0.008	0.007	0.006	0.003
Panel C. Higher Income (>138% FPL)					
Pregnancy Dental Benefit	0.0588***	0.0115	-0.0273**	0.00176	0.00396***
	(0.0112)	(0.0145)	(0.0126)	(0.00368)	(0.00124)
Baseline mean	0.299	0.111	0.106	0.092	0.015
Wild cluster bootstrap p-value	0.035	0.628	0.172	0.712	0.015
Observations	21721	19636	20650	21854	21854
R-squared	0.024	0.010	0.017	0.006	0.005

Notes: Data come from the Pregnancy Risk Assessment Monitoring System (PRAMS), 2012-2019. This table reports estimates of the effect of Medicaid pregnancy dental benefits on prenatal dental cleaning (Column (1)) and birth outcomes (Columns (2)-(5)), excluding states that offer pregnancy-only dental coverage (UT and VA). Estimates are shown for the full sample as well as stratified by income relative to 138 percent of the federal poverty level, a threshold commonly used to define Medicaid eligibility for nonpregnant adults. Results are estimated using a stacked difference-in-differences framework that accounts for staggered policy adoption across states. All models control for maternal demographic characteristics, household income, and pre-pregnancy risk factors, along with state-level socioeconomic conditions, Medicaid eligibility limits, managed care penetration, and healthcare infrastructure characteristics. Specifications include state and birth quarter-by-year fixed effects and apply PRAMS survey weights. Conventional standard errors clustered at the state level are reported in parentheses, and statistical significance stars are based on these standard errors (*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$). Wild cluster bootstrap p-values clustered at the state level using Webb weights are reported in a separate row to provide inference robust to the limited number of state clusters.

Table 5: Stacked DiD Estimates of the Effect of Medicaid Pregnancy and Pre-Conception Dental Benefits on Prenatal Dental Cleanings and Birth Outcomes, 2012-2019 PRAMS

	(1) Dental Cleaning	(2) Small for Gestational Age (P10)	(3) Preterm Birth	(4) Low Birth Weight	(5) Very Low Birth Weight
Pregnancy Dental Benefit	0.0877***	-0.0208***	-0.00569	-0.00455	-0.00477***
	(0.0103)	(0.00614)	(0.00612)	(0.00273)	(0.00135)
Non-Pregnant Medicaid Expansion	0.0187	-0.00287	-0.0210*	-0.00347	0.000106
	(0.0141)	(0.00854)	(0.0119)	(0.00498)	(0.00166)
Baseline mean	0.255	0.122	0.117	0.107	0.018
Wild cluster bootstrap p (Pregnancy)	0.011	0.319	0.504	0.090	0.097
Wild cluster bootstrap p (Non-pregnancy)	0.315	0.827	0.282	0.675	0.966
Observations	94612	86525	90604	95130	95130
R-squared	0.035	0.013	0.024	0.022	0.009

Notes: Data come from the Pregnancy Risk Assessment Monitoring System (PRAMS), 2012-2019. This table reports stacked difference-in-differences estimates of the effect of Medicaid pregnancy dental benefits on prenatal dental cleaning (Column (1)) and birth outcomes (Columns (2)-(5)) while controlling for adult Medicaid dental coverage in the pre-conception period. Pre-pregnancy dental coverage is defined as the availability of adult Medicaid dental benefits up to 1.5 years prior to birth. Models include maternal demographic characteristics, state-level socioeconomic and healthcare controls, and state and birth year fixed effects. All models incorporate PRAMS survey weights and include state and birth quarter-by-year fixed effects. Conventional standard errors clustered at the state level are reported in parentheses, and statistical significance stars are based on these standard errors (*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$). Wild cluster bootstrap p-values clustered at the state level using Webb weights are reported in a separate row to provide inference robust to the limited number of state clusters.

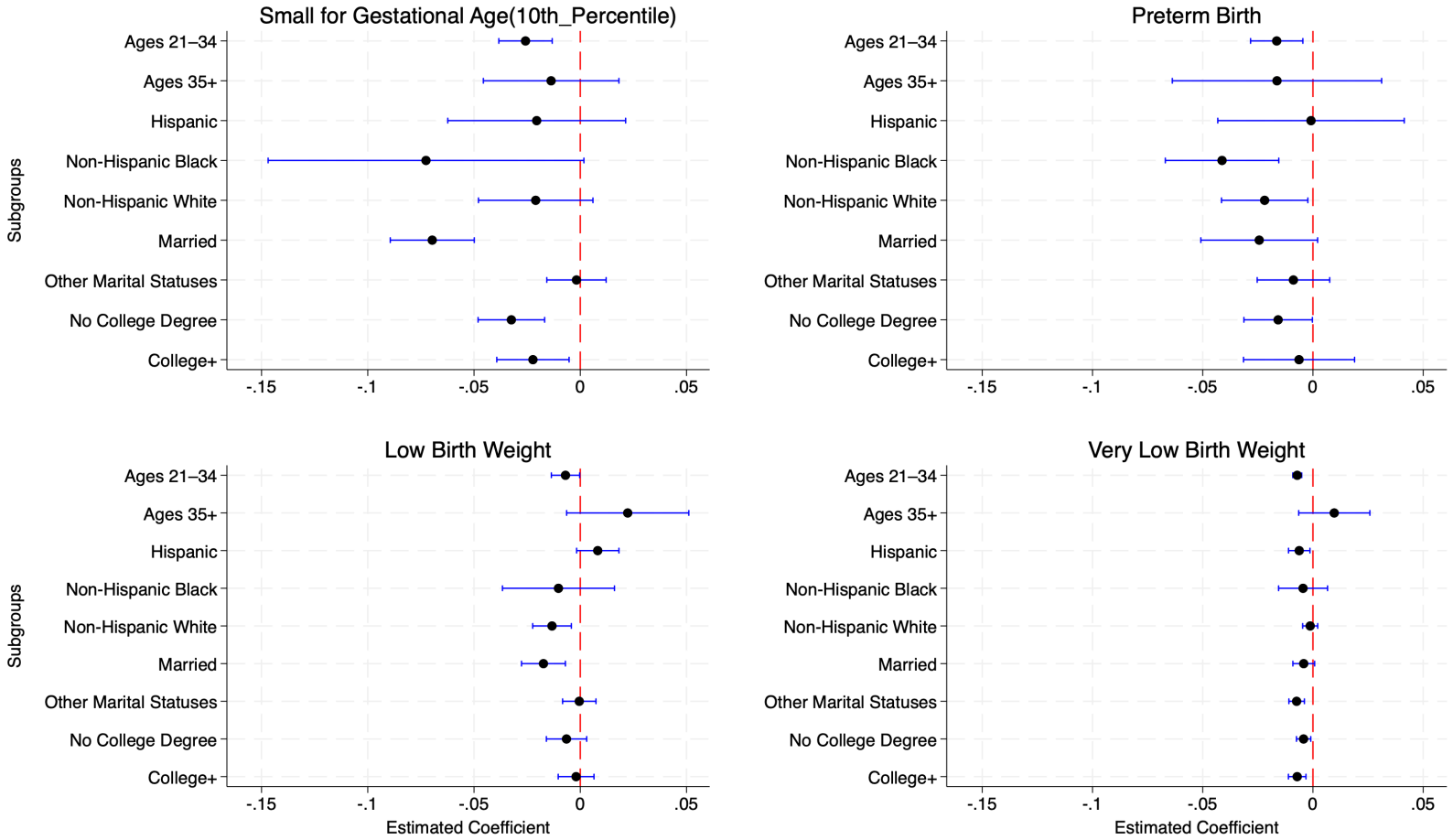
Table 6: Stacked DiD Estimates of the Effect of Medicaid Pregnancy Dental Benefits on Dental Insurance, Oral Health Literacy, and Maternal Health Behaviors

	(1) Has Dental Insurance	(2) Received Dental Care Advice	(3) Knows Dental Care is Important	(4) Adequate Prenatal Care (Kotelchuck)	(5) Maternal Smoking (Any)	(6) Smoking Last 3 Months of Pregnancy	(7) Non-Smoking After Pregnancy
Pregnancy Dental Benefit	0.135***	0.0576**	0.0113	0.000770	-0.0149*	-0.0121	0.0124
	(0.0259)	(0.0214)	(0.0202)	(0.0181)	(0.00805)	(0.0152)	(0.00814)
Baseline mean	0.333	0.378	0.854	0.887	0.196	0.224	0.290
Wild cluster bootstrap p-value	0.045	0.067	0.750	0.981	0.512	0.679	0.252
Observations	47533	48298	48269	75602	93584	93937	93870
R-squared	0.113	0.031	0.042	0.212	0.165	0.162	0.159

Notes: Data come from the Pregnancy Risk Assessment Monitoring System (PRAMS), 2012-2019. This table reports estimates of the effects of Medicaid pregnancy dental benefits on prenatal care adequacy, dental insurance coverage, oral health awareness, provider dental advice, and maternal smoking behaviors. Each column presents results from a separate regression estimated using a stacked difference-in-differences framework that accounts for staggered policy adoption. Models control for maternal demographics (age, race/ethnicity, education, marital status, and urban/rural residence), household income, and pre-pregnancy risk factors, as well as state-level socioeconomic conditions, Medicaid eligibility limits, managed care penetration, and healthcare infrastructure characteristics. All models include state and birth quarter-by-year fixed effects and incorporate PRAMS survey weights. Conventional standard errors clustered at the state level are reported in parentheses, and statistical significance stars are based on these standard errors (*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$). Wild cluster bootstrap p-values clustered at the state level using Webb weights are reported in a separate row to provide inference robust to the limited number of state clusters.

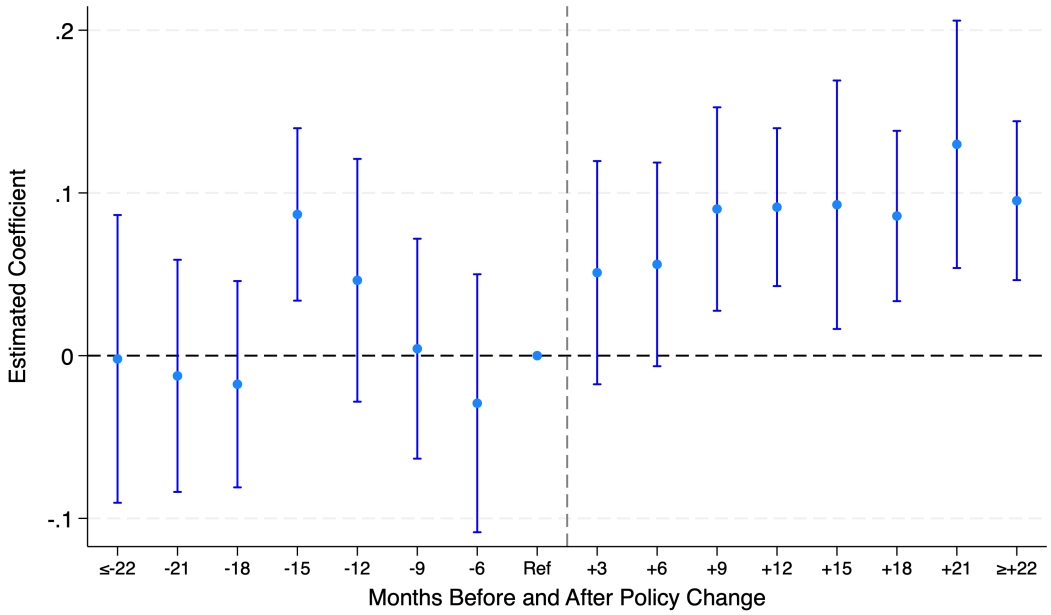
Appendix

Figure A.1: Heterogeneous Effects of Medicaid Pregnancy Dental Benefits on Birth Outcomes by Subgroup, 2012-2019 PRAMS



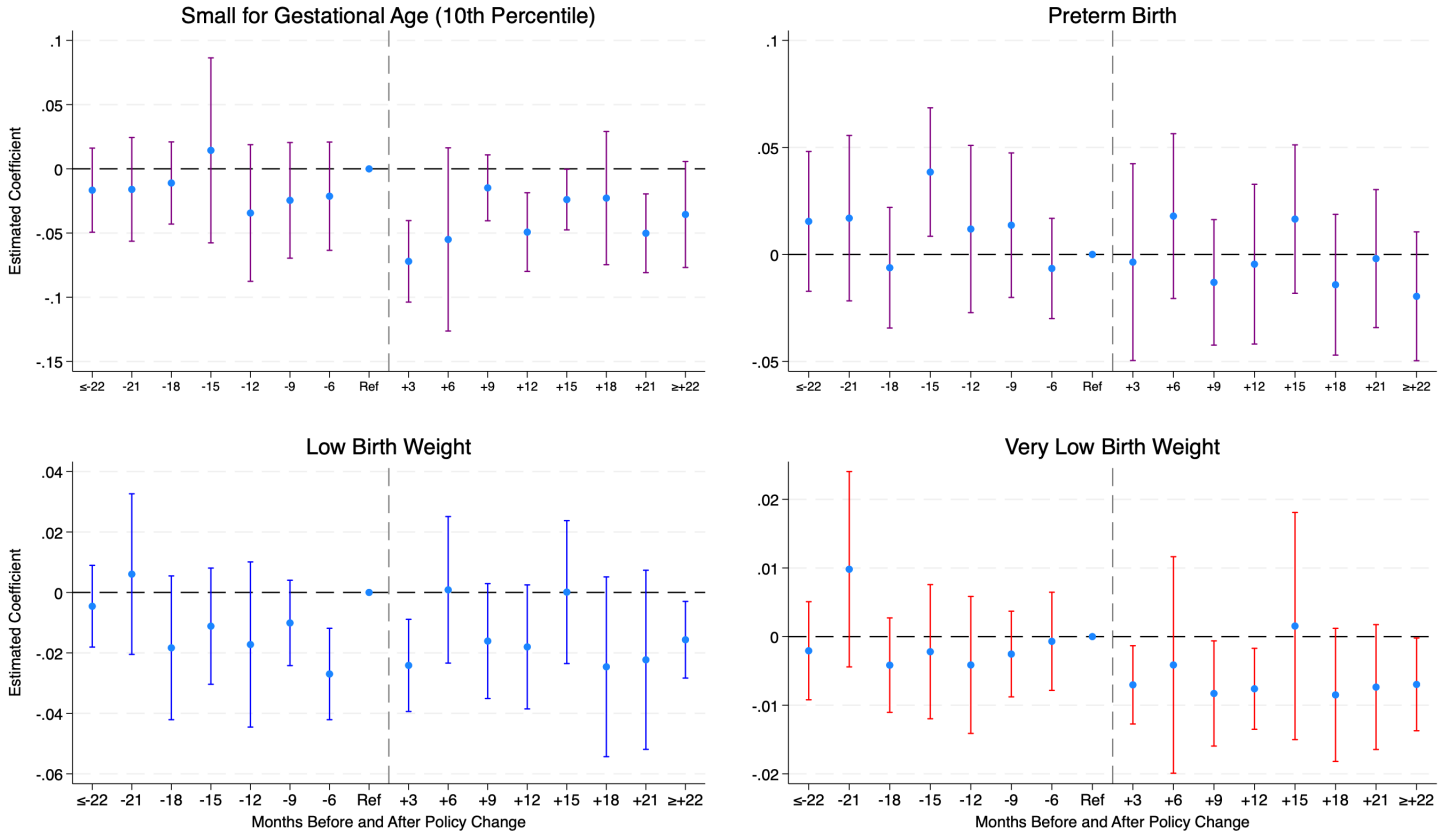
Notes: Data come from the Pregnancy Risk Assessment Monitoring System (PRAMS), 2012-2019. This figure presents heterogeneous effects of Medicaid pregnancy dental benefits on four birth outcomes: small for gestational age (10th percentile), preterm birth, low birth weight, and very low birth weight. Estimates are shown for demographic subgroups, including maternal age (21-25, 26-34, 35+), race/ethnicity (Hispanic, Non-Hispanic Black, Non-Hispanic White), marital status (married vs. other), and education (no college degree vs. college+). Each point represents the estimated coefficient from a separate subgroup regression, and the horizontal bars indicate 95% confidence intervals. The vertical red line at zero represents no effect.

Figure A.2: Event Study Estimates of the Effect of Medicaid Pregnancy Dental Benefits on Prenatal Dental Cleanings, Including Post-COVID-19 Period, 2012-2021 PRAMS



Notes: Data come from the Pregnancy Risk Assessment Monitoring System (PRAMS), 2012-2021. The figure displays event study estimates of the effect of Medicaid pregnancy dental benefit expansions on prenatal dental cleaning rates, estimated using a stacked difference-in-differences framework with data extending through the post-COVID-19 period. Models control for maternal demographics (age, race/ethnicity, education, marital status, urban/rural residence, pre-pregnancy risk factors, and household income), state-level socioeconomic and healthcare characteristics (Medicaid eligibility limits, ACA expansion, managed care penetration, dentist supply, unemployment rate, health centers per capita, and fertility rate), and COVID-19-related measures (case and death rates, business closures, stay-at-home orders, elective care restrictions, and overall government response index). The x-axis represents time in months before and after policy adoption, and the y-axis shows the estimated coefficients. 95% confidence intervals are indicated by the vertical lines around the point estimates. The reference group is 0-3 months before policy implementation. These estimates test the parallel trends assumption and capture dynamic treatment effects over time.

Figure A.3: Event Study Estimates of the Effect of Medicaid Pregnancy Dental Benefits on Birth Outcomes, Including Post-COVID-19 Period, 2012-2021 PRAMS



Notes: Data come from the Pregnancy Risk Assessment Monitoring System (PRAMS), 2012-2021. The figure presents event study estimates of the effect of Medicaid pregnancy dental benefits on key birth outcomes: small for gestational age (10th percentile), preterm birth, low birth weight, and very low birth weight, using a stacked difference-in-differences framework with data extending through the post-COVID-19 period. Each panel corresponds to a separate outcome. The x-axis represents time in months before and after policy adoption, and the y-axis shows the estimated coefficients. 95% confidence intervals are indicated by the vertical lines around the point estimates. Models control for maternal demographic characteristics, state-level socioeconomic and healthcare factors, and COVID-19-related measures (case and death rates, business closures, stay-at-home orders, elective care restrictions, and overall government response index). The reference group is 0-3 months before policy implementation. These estimates assess the parallel trends assumption and capture dynamic treatment effects over time.

Table A.1: Balance Test: Covariate Differences by Medicaid Pregnancy Dental Benefit Status, 2012–2019 PRAMS

	(1) Low Education	(2) High Education	(3) Age (Years)	(4) Hispanic	(5) Non-Hispanic Black	(6) Non-Hispanic White	(7) Non-Hispanic Other	(8) Married	(9) Other Marital Status
Pregnancy Dental Benefit	0.00669	-0.00669	-0.175	-0.0449***	0.0649*	0.0231	-0.0431	-0.0237**	0.0237**
	(0.0319)	(0.0319)	(0.192)	(0.00874)	(0.0351)	(0.0443)	(0.0636)	(0.00913)	(0.00913)
Wild cluster bootstrap p-value	0.918	0.918	0.596	0.033	0.225	0.735	0.679	0.208	0.208
Observations	95142	95142	95142	95142	95142	95142	95142	95142	95142
R-squared	0.010	0.010	0.024	0.093	0.118	0.102	0.347	0.027	0.027

	(10) < $20,000 Income	(11) $20,000–$33,000 Income	(12) $33,000–$55,000 Income	(13) $55,000–$80,000 Income	(14) > $80,000 Income	(15) Missing Income	(16) Prepregnancy Risk	(17) No Risk	(18) Missing Risk Info	(19) Urban Areas	(20) Rural Areas
Pregnancy Dental Benefit	-0.0213	0.00770	-0.0217	0.0141	0.00317	0.0141	0.106*	0.0170*	-0.122*	0.0382	-0.0382
	(0.0228)	(0.00839)	(0.0155)	(0.00874)	(0.00459)	(0.00874)	(0.0573)	(0.00897)	(0.0655)	(0.0283)	(0.0283)
Wild cluster bootstrap p-value	0.625	0.432	0.547	0.392	0.649	0.392	0.316	0.197	0.294	0.370	0.370
Observations	95142	95142	95142	95142	95142	95142	95142	95142	95142	95142	95142
R-squared	0.067	0.014	0.038	0.035	0.004	0.035	0.440	0.017	0.816	0.109	0.109

Notes: Data come from the Pregnancy Risk Assessment Monitoring System (PRAMS), 2012–2019. Each coefficient is obtained from a separate stacked difference-in-differences regression in which the listed covariate is the dependent variable and is regressed on an indicator for Medicaid pregnancy dental benefit adoption. All models control for state-level socioeconomic and healthcare covariates, include state and year fixed effects, and incorporate PRAMS survey weights. Standard errors clustered at the state level are reported in parentheses. Statistical significance stars are based on conventional clustered standard errors (*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$). Wild cluster bootstrap p-values clustered at the state level using Webb weights are reported separately to provide inference robust to the limited number of state clusters.

Table A.2: Stacked DiD Estimates of the Effect of Medicaid Pregnancy Dental Benefits on Prenatal Dental Cleanings, Varying Controls, 2012-2019 PRAMS

	(1)	(2)	(3)	(4)
	Dental Cleaning			
Pregnancy Dental Benefit	0.0822***	0.0792***	0.0805***	0.0949***
	(0.0215)	(0.0196)	(0.00977)	(0.0105)
State	YES	YES	YES	YES
Year	YES	YES	YES	YES
Demographics		YES	YES	YES
Contextual Factors			YES	YES
Health Resources				YES
Wild cluster bootstrap p-value	0.015	0.048	0.001	0.002
Observations	96422	94612	94612	94612
R-squared	0.015	0.033	0.035	0.035

Notes: Data come from the Pregnancy Risk Assessment Monitoring System (PRAMS), 2012-2019. The table reports estimates of the effect of Medicaid pregnancy dental benefits on prenatal dental cleaning using a stacked difference-in-differences design. Each column corresponds to a separate regression that sequentially adds controls for maternal demographics, state-level socioeconomic characteristics, and healthcare resource availability. Contextual factors include state fertility rates, unemployment rates, managed care organization penetration, and federally qualified health centers per capita. Health resources include ACA Medicaid expansion status and dentists per capita. All models include state and birth quarter-by-year fixed effects and apply PRAMS survey weights to ensure population representativeness. Conventional standard errors clustered at the state level are reported in parentheses, and statistical significance stars are based on these standard errors (*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$). Wild cluster bootstrap p-values clustered at the state level using Webb weights are reported in a separate row to provide inference robust to the limited number of state clusters.

Table A.3: Two-way-fixed-effects Estimates of the Effect of Medicaid Non-Pregnancy Dental Benefits on Self-Reported General Health, 2000-2022 BRFSS

	(1) Good and above	(2) Very good and above	(3) Excellent
Non-Pregnancy Dental Benefit	0.013**	0.008	0.012**
	(0.006)	(0.009)	(0.005)
Observations	155,746	155,746	155,746
R-squared	0.072	0.084	0.024

Notes: Data come from the Behavioral Risk Factor Surveillance System (BRFSS), 2000-2022. We include all available years and states to maximize statistical power. The table presents estimates of the effect of adult Medicaid dental expansions on self-reported general health among non-pregnant reproductive-age women. The outcome variables are indicators for self-reported health status being good or better (Column (1)), very good or better (Column (2)), and excellent (Column (3)). Individual and family-level controls include age fixed effects, income level, highest educational attainment, race and ethnicity, marital status, and the number of adults and children in the household. State-year-level controls include an ACA expansion indicator, managed care penetration rate, the number of Federally Qualified Health Centers per capita, and the number of dentists per capita. During the COVID-19 pandemic, additional controls include an indicator for stay-at-home orders, an indicator for elective procedure delay orders, the log of yearly total new cases, and the log of yearly total deaths. All specifications include state and year fixed effects. Conventional standard errors clustered at the state level are reported in parentheses, and statistical significance stars are based on these standard errors (*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$).

Table A.4: Stacked DiD Estimates of the Effect of Medicaid Pregnancy Dental Benefits on Prenatal Dental Cleanings and Birth Outcomes, Including Post-COVID-19 Period, 2012-2021 PRAMS

	(1) Dental Cleaning	(2) Small for Gestational Age (P10)	(3) Preterm Birth	(4) Low Birth Weight	(5) Very Low Birth Weight
Pregnancy Dental Benefit	0.0789***	-0.0267***	-0.0119***	-0.00293	-0.00484***
	(0.0110)	(0.00604)	(0.00435)	(0.00299)	(0.00108)
Baseline Mean	0.251	0.121	0.119	0.108	0.018
Wild cluster bootstrap p-value	0.001	0.138	0.056	0.432	0.068
Observations	128268	116533	121930	128824	128824
R-squared	0.035	0.013	0.025	0.022	0.010

Notes: Data come from the Pregnancy Risk Assessment Monitoring System (PRAMS), 2012-2021. The table presents estimates of the effect of Medicaid pregnancy dental benefits on prenatal dental cleaning (Column (1)) and birth outcomes (Columns (2)-(5)) using data that include the post-COVID-19 period. Each column reports results from a separate regression controlling for maternal demographics (age, race/ethnicity, education, marital status, urban/rural residence, pre-pregnancy risk factors), and state-level socioeconomic and healthcare characteristics (Medicaid eligibility limits, managed care penetration, dentist supply, unemployment rate, health centers per capita, and fertility rate). Models also adjust for COVID-19-related factors, including case and death rates, business closures, stay-at-home orders, elective care restrictions, and the government response index. All models include state and birth quarter-by-year fixed effects and apply PRAMS survey weights to ensure population representativeness. Conventional standard errors clustered at the state level are reported in parentheses, and statistical significance stars are based on these standard errors (*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$). Wild cluster bootstrap p-values clustered at the state level using Webb weights are reported in a separate row to provide inference robust to the limited number of state clusters.

Table A.5: Stacked DiD Estimates of the Effect of Medicaid Pregnancy Dental Benefits on Prenatal Dental Cleanings and Birth Outcomes, Varying Event Windows, 2012-2019 PRAMS

	(1) Dental Cleaning	(2) Small for Gestational Age (P10)	(3) Preterm Birth	(4) Low Birth Weight	(5) Very Low Birth Weight
24-Month Window					
Pregnancy Dental Benefit	0.0925***	-0.0182**	-0.00749	-0.000438	-0.00412***
	(0.0102)	(0.00675)	(0.00806)	(0.00228)	(0.00114)
Baseline Mean	0.266	0.120	0.113	0.103	0.018
Wild cluster bootstrap p-value	0.010	0.314	0.487	0.877	0.041
Observations	62472	58827	61719	62870	62870
R-squared	0.039	0.015	0.021	0.022	0.008
36-Month Window					
Pregnancy Dental Benefit	0.0832***	-0.0207***	-0.0141**	-0.000113	-0.00502***
	(0.00902)	(0.00575)	(0.00671)	(0.00245)	(0.00129)
Baseline Mean	0.261	0.122	0.114	0.105	0.017
Wild cluster bootstrap p-value	0.003	0.216	0.171	0.980	0.034
Observations	80128	74117	77670	80617	80617
R-squared	0.038	0.014	0.022	0.022	0.009
60-Month Window					
Pregnancy Dental Benefit	0.125***	-0.0233***	-0.00529	0.00121	-0.00183
	(0.0170)	(0.00541)	(0.00543)	(0.00228)	(0.00170)
Baseline Mean	0.250	0.122	0.118	0.107	0.018
Wild cluster bootstrap p-value	0.004	0.096	0.562	0.599	0.559
Observations	108518	99305	103974	109043	109043
R-squared	0.035	0.013	0.025	0.022	0.010

Notes: Data come from the Pregnancy Risk Assessment Monitoring System (PRAMS), 2012-2019. The table presents stacked difference-in-differences (SDiD) estimates of the effect of Medicaid pregnancy dental benefits on prenatal dental cleaning (Column (1)) and birth outcomes (Columns (2)-(5)) using alternative event window lengths of 24, 36, and 60 months. Each panel corresponds to a distinct specification defining the pre- and post-policy observation periods used to construct the stacked DiD estimators. All models control for maternal demographic characteristics (age, race/ethnicity, education, marital status, urban/rural residence, and pre-pregnancy risk factors) and state-level socioeconomic and healthcare variables (Medicaid eligibility limits, managed care penetration, dentist supply, unemployment rate, fertility rate, and health centers per capita). Models include state-by-sub-experiment and birth quarter-by-year fixed effects and are weighted using PRAMS survey weights. Conventional standard errors clustered at the state level are reported in parentheses, and statistical significance stars are based on these standard errors (*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$). Wild cluster bootstrap p-values clustered at the state level using Webb weights are reported in a separate row to provide inference robust to the limited number of state clusters.

Table A.6: Stacked DiD Estimates of the Effect of Medicaid Pregnancy Dental Benefits on Prenatal Dental Cleanings and Birth Outcomes, Alternative Medicaid Sample Definitions, 2012-2019 PRAMS

	(1) Dental Cleaning	(2) Small for Gestational Age (P10)	(3) Preterm Birth	(4) Low Birth Weight	(5) Very Low Birth Weight
Panel A. Birth Certificate Medicaid or Self-Report When BC Missing					
Pregnancy Dental Benefit	0.0949***	-0.0219***	-0.0135***	-0.00590*	-0.00473***
	(0.0105)	(0.00568)	(0.00399)	(0.00302)	(0.00109)
Baseline Mean	0.255	0.122	0.117	0.107	0.018
Wild cluster bootstrap p-value	0.002	0.230	0.036	0.064	0.079
Observations	94612	86525	90604	95130	95130
R-squared	0.035	0.013	0.024	0.022	0.009
Panel B. Birth Certificate Medicaid Required					
Pregnancy Dental Benefit	0.103***	-0.0314***	-0.0142***	-0.00371	-0.00343**
	(0.0124)	(0.00737)	(0.00436)	(0.00285)	(0.00130)
Baseline mean	0.253	0.121	0.118	0.108	0.018
Wild cluster bootstrap p-value	0.018	0.234	0.063	0.385	0.247
Observations	84797	77365	80987	85302	85302
R-squared	0.035	0.013	0.024	0.022	0.010
Panel C. Birth Certificate and Self-Reported Medicaid Required					
Pregnancy Dental Benefit	0.102***	-0.0330***	-0.0212***	-0.00123	-0.00512***
	(0.0111)	(0.00811)	(0.00550)	(0.00249)	(0.00113)
Baseline mean	0.240	0.122	0.119	0.111	0.018
Wild cluster bootstrap p-value	0.023	0.175	0.069	0.861	0.019
Observations	73220	67569	70729	73643	73643
R-squared	0.035	0.016	0.026	0.023	0.010
Panel D. Birth Certificate or Self-Reported Medicaid					
Pregnancy Dental Benefit	0.0943***	-0.0109**	-0.0107**	-0.00452	-0.00354***
	(0.0108)	(0.00522)	(0.00446)	(0.00417)	(0.00109)
Baseline mean	0.260	0.120	0.116	0.106	0.017
Wild cluster bootstrap p-value	0.003	0.564	0.078	0.276	0.123
Observations	104460	95397	99944	104991	104991
R-squared	0.034	0.012	0.021	0.021	0.008

Notes: Data come from the Pregnancy Risk Assessment Monitoring System (PRAMS), 2012-2019. This table reports stacked difference-in-differences estimates of the effect of Medicaid pregnancy dental benefits on prenatal dental cleaning (Column (1)) and birth outcomes (Columns (2)-(5)) using alternative definitions of Medicaid coverage constructed from birth certificate records and maternal self-reports. Estimates are derived from specifications that account for staggered policy adoption across states. All models control for maternal demographic characteristics (age, race/ethnicity, education, marital status, urban or rural residence, and pre-pregnancy risk factors) and state-level socioeconomic and healthcare characteristics (Medicaid eligibility limits, managed care penetration, dentist supply, unemployment rate, fertility rate, and health centers per capita). Models include state-by-sub-experiment and birth quarter-by-year fixed effects and are weighted using PRAMS survey weights. Conventional standard errors clustered at the state level are reported in parentheses, and statistical significance stars are based on these standard errors (*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$). Wild cluster bootstrap p-values clustered at the state level using Webb weights are reported in a separate row to provide inference robust to the limited number of state clusters.

Table A.7: Stacked DiD Estimates of the Effect of Medicaid Pregnancy Dental Benefits on Prenatal Dental Care and Birth Outcomes by Pre-Conception Medicaid Eligibility, 2012-2019 PRAMS

	(1) Dental Cleaning	(2) Small for Gestational Age (P10)	(3) Preterm Birth	(4) Low Birth Weight	(5) Very Low Birth Weight
Panel A. Any Pregnancy Exposure vs None					
Any exposure (any trimester)	0.0975***	-0.0170***	0.00314	0.0123	0.000700
	(0.0121)	(0.00631)	(0.00921)	(0.00753)	(0.00193)
Baseline mean	0.260	0.122	0.116	0.107	0.018
Wild cluster bootstrap p-value	0.014	0.345	0.814	0.402	0.723
Observations	90174	86403	90604	90597	90597
R-squared	0.034	0.013	0.024	0.023	0.010
Panel B. Early Exposure (Full or First Trimester) vs All Others					
Early exposure (Full, 1st or 2nd trimester)	0.0935***	-0.0189***	-0.000427	0.00579	0.00198
	(0.0136)	(0.00583)	(0.00741)	(0.00465)	(0.00212)
Baseline mean	0.260	0.122	0.116	0.107	0.018
Wild cluster bootstrap p-value	0.012	0.203	0.959	0.472	0.429
Observations	90174	86403	90604	90597	90597
R-squared	0.034	0.013	0.024	0.023	0.010
Panel C. Percent of Pregnancy Exposed					
Exposure intensity (percent of pregnancy)	0.110***	-0.0161***	-0.00501	0.00744	0.00124
	(0.0129)	(0.00594)	(0.00823)	(0.00626)	(0.00226)
Baseline mean	0.260	0.122	0.116	0.107	0.018
Wild cluster bootstrap p-value	0.006	0.276	0.569	0.511	0.617
Observations	90174	86403	90604	90597	90597
R-squared	0.034	0.013	0.024	0.023	0.010

Notes: Data come from the Pregnancy Risk Assessment Monitoring System (PRAMS), 2012-2019. This table reports intent-to-treat estimates of the effect of Medicaid pregnancy dental benefits on prenatal dental cleaning (Column (1)) and birth outcomes (Columns (2)-(5)) using an imputed conception date. Conception timing is constructed by accounting for gestational length, allowing treatment exposure to be measured at the week level over the course of pregnancy. Estimates are derived using a stacked difference-in-differences framework that accounts for staggered policy adoption across states. All models include maternal demographic characteristics, state-level socioeconomic controls, health system variables, and state and quarter-by-year fixed effects. Regressions are weighted using PRAMS survey weights to ensure population representativeness. Conventional standard errors clustered at the state level are reported in parentheses, and statistical significance stars are based on these standard errors (*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$). Wild cluster bootstrap p-values clustered at the state level using Webb weights are reported in a separate row to provide inference robust to the limited number of state clusters.

Table A.8: Stacked DiD Estimates of the Effects of Medicaid Pregnancy Dental Benefits on Prenatal Dental Cleanings and Birth Outcomes, Leave-One-Out Estimates, 2012-2019 PRAMS

	(1) Full Sample	(2) Excl. UT	(3) Excl. IL	(4) Excl. CO	(5) Excl. VA	(6) Excl. SC
Panel A: Dental Cleaning						
Pregnancy Dental Benefit	0.0949***	0.0965***	0.109***	0.0915***	0.0975***	0.0860***
	(0.0105)	(0.0109)	(0.0175)	(0.0110)	(0.0117)	(0.00887)
Baseline Mean	0.255	0.253	0.256	0.255	0.255	0.254
Wild cluster bootstrap p-value	0.002	0.001	0.029	0.001	0.027	0.001
Observations	94612	77411	74728	75688	74517	76104
R-squared	0.035	0.032	0.039	0.035	0.035	0.035
Panel B: Small for Gestational Age (P10)						
Pregnancy Dental Benefit	-0.0219***	-0.0234***	-0.0236**	-0.0200***	-0.0265***	-0.0202***
	(0.00568)	(0.00607)	(0.0107)	(0.00707)	(0.00623)	(0.00554)
Baseline Mean	0.122	0.122	0.122	0.121	0.122	0.122
Wild cluster bootstrap p-value	0.230	0.174	0.414	0.421	0.227	0.184
Observations	86525	70689	68298	69179	68210	69724
R-squared	0.013	0.013	0.013	0.013	0.013	0.013
Panel C: Preterm Birth						
Pregnancy Dental Benefit	-0.0135***	-0.0116**	-0.00914	-0.0152***	-0.0153***	-0.0151***
	(0.00399)	(0.00448)	(0.00685)	(0.00362)	(0.00363)	(0.00519)
Baseline Mean	0.117	0.117	0.117	0.117	0.117	0.116
Wild cluster bootstrap p-value	0.036	0.104	0.332	0.009	0.017	0.071
Observations	90604	74003	71515	72407	71502	72989
R-squared	0.024	0.024	0.024	0.024	0.023	0.023
Panel D: Low Birth Weight						
Pregnancy Dental Benefit	-0.00590*	-0.00656*	-0.0104**	-0.00546	-0.00457	-0.00351
	(0.00302)	(0.00337)	(0.00466)	(0.00329)	(0.00282)	(0.00242)
Baseline Mean	0.107	0.107	0.107	0.107	0.107	0.107
Wild cluster bootstrap p-value	0.064	0.028	0.141	0.074	0.279	0.204
Observations	95130	77826	75143	76107	74929	76515
R-squared	0.022	0.022	0.021	0.022	0.022	0.022
Panel E: Very Low Birth Weight						
Pregnancy Dental Benefit	-0.00473***	-0.00538***	-0.00393*	-0.00445***	-0.00422**	-0.00523***
	(0.00109)	(0.00102)	(0.00226)	(0.00112)	(0.00160)	(0.000986)
Baseline Mean	0.018	0.018	0.017	0.018	0.018	0.017
Wild cluster bootstrap p-value	0.079	0.036	0.354	0.174	0.294	0.019
Observations	95130	77826	75143	76107	74929	76515
R-squared	0.009	0.009	0.010	0.010	0.009	0.009

Notes: Data come from the Pregnancy Risk Assessment Monitoring System (PRAMS), 2012-2019. The table presents leave-one-out stacked difference-in-differences estimates of the effect of Medicaid pregnancy dental benefits on prenatal dental cleaning (Panel (A)) and birth outcomes (Panels (B)-(D)). Each column excludes one treated state (Utah, Illinois, Colorado, Virginia, or South Carolina) from the estimation sample to assess the influence of individual states on overall results. Models control for maternal demographic characteristics (age, race/ethnicity, education, marital status, urban/rural residence, and pre-pregnancy risk factors) and state-level socioeconomic and healthcare characteristics (Medicaid eligibility limits, managed care penetration, dentist supply, unemployment rate, fertility rate, and health centers per capita). All specifications include state-by-sub-experiment and birth quarter-by-year fixed effects and apply PRAMS survey weights. Conventional standard errors clustered at the state level are reported in parentheses, and statistical significance stars are based on these standard errors (*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$). Wild cluster bootstrap p-values clustered at the state level using Webb weights are reported in a separate row to provide inference robust to the limited number of state clusters.

Table A.9: Two-way-fixed-effects Estimates of the Effect of Medicaid Pregnancy Dental Benefits on Prenatal Dental Cleanings and Birth Outcomes, 2012-2019 PRAMS

	(1) Dental Cleaning	(2) Small for Gestational Age (P10)	(3) Preterm Birth	(4) Low Birth Weight	(5) Very Low Birth Weight
Pregnancy Dental Benefit	0.106***	-0.00973	-0.0106	-0.0112**	-0.00393
	(0.0143)	(0.00953)	(0.00667)	(0.00504)	(0.00247)
Baseline Mean	0.252	0.121	0.117	0.105	0.018
Wild cluster bootstrap p-value	0.004	0.382	0.237	0.145	0.397
Observations	31370	29115	30381	31483	31483
R-squared	0.034	0.012	0.024	0.020	0.010

Notes: Data come from the Pregnancy Risk Assessment Monitoring System (PRAMS), 2012-2019. The table presents estimates of the effect of Medicaid pregnancy dental benefits on prenatal dental cleaning (Column (1)) and birth outcomes (Columns (2)-(5)) using a traditional two-way fixed effects specification. Each column corresponds to results from a separate regression, with robust standard errors clustered at the state level shown in parentheses. The baseline mean is the average rate of each outcome in states without Medicaid pregnancy dental benefits. Covariates include maternal age, race/ethnicity, education, marital status, urban/rural residence, and pre-pregnancy risk factors, along with state-level controls such as Medicaid eligibility limits, managed care penetration, and broader contextual factors. All models include state and birth quarter-by-year fixed effects to account for unobserved heterogeneity across states and over time. Conventional standard errors clustered at the state level are reported in parentheses, and statistical significance stars are based on these standard errors (*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$). Wild cluster bootstrap p-values clustered at the state level using Webb weights are reported in a separate row to provide inference robust to the limited number of state clusters.

Table A.10: Stacked DiD Estimates of the Effects of Medicaid Pregnancy Dental Benefits, Intent-to-Treat Estimates, 2012-2019 PRAMS

	(1) All Sample	(2) Low Education (≤ Associate)	(3) High Education (≥ Bachelor's)	(4) Income ≤ $50k	(5) Income ≥ $75k
Panel A: Dental Cleaning					
Pregnancy Dental Benefit	0.0430***	0.0532***	0.0424***	0.0582***	-0.00167
	(0.00884)	(0.0112)	(0.0109)	(0.0118)	(0.0114)
Baseline mean	0.415	0.261	0.511	0.291	0.699
Wild cluster bootstrap p-value	0.009	0.013	0.006	0.005	0.957
Observations	199716	75316	124400	115381	46336
R-squared	0.148	0.048	0.129	0.040	0.053
Panel B: Small for Gestational Age (P10)					
Pregnancy Dental Benefit	-0.00527	-0.00403	-0.00829	-0.0122**	-0.00510
	(0.00450)	(0.00721)	(0.00613)	(0.00530)	(0.0107)
Baseline mean	0.101	0.124	0.087	0.114	0.076
Wild cluster bootstrap p-value	0.619	0.694	0.529	0.430	0.709
Observations	181122	69378	111744	105393	41423
R-squared	0.012	0.016	0.010	0.014	0.013
Panel C: Preterm Birth					
Pregnancy Dental Benefit	-0.000970	-0.00843	0.00236	-0.00149	0.00179
	(0.00244)	(0.00883)	(0.00524)	(0.00392)	(0.00571)
Baseline mean	0.099	0.119	0.087	0.110	0.075
Wild cluster bootstrap p-value	0.799	0.583	0.759	0.776	0.730
Observations	190768	72343	118425	110355	44264
R-squared	0.020	0.022	0.018	0.020	0.018
Panel D: Low Birth Weight					
Pregnancy Dental Benefit	0.000430	-0.00199	0.000842	0.000147	0.000334
	(0.00218)	(0.00321)	(0.00300)	(0.00271)	(0.00502)
Baseline mean	0.083	0.103	0.071	0.099	0.053
Wild cluster bootstrap p-value	0.846	0.719	0.849	0.951	0.957
Observations	200361	75681	124680	115806	46386
R-squared	0.022	0.021	0.019	0.020	0.010
Panel E: Very Low Birth Weight					
Pregnancy Dental Benefit	-0.00124*	-0.000972	-0.00133	-0.00226*	0.00211
	(0.000692)	(0.00128)	(0.00115)	(0.00119)	(0.00161)
Baseline mean	0.014	0.017	0.011	0.017	0.008
Wild cluster bootstrap p-value	0.231	0.639	0.551	0.286	0.663
Observations	200361	75681	124680	115806	46386
R-squared	0.008	0.009	0.007	0.008	0.008

Notes: Data come from the Pregnancy Risk Assessment Monitoring System (PRAMS), 2012-2019. This table reports intent-to-treat estimates of the effect of Medicaid pregnancy dental benefits on prenatal dental cleaning (Panel (A)) and birth outcomes (Panels (B)-(D)), stratified by education and income subgroups. Low education is defined as an associate degree or less; high education is defined as a bachelor's degree or higher. Income categories are based on midpoint estimates of self-reported household income brackets. Estimates are derived using a stacked difference-in-differences framework that accounts for staggered policy adoption. All models include individual demographic covariates, state-level socioeconomic controls, state and birth quarter-by-year fixed effects, and incorporate PRAMS survey weights to ensure population representativeness. Conventional standard errors clustered at the state level are reported in parentheses, and statistical significance stars are based on these standard errors (*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$). Wild cluster bootstrap p-values clustered at the state level using Webb weights are reported in a separate row to provide inference robust to the limited number of state clusters.